

Article

A Per-Pixel Provenance Contract and Evidence Profile for Cross-Rig Camera and LiDAR Reprojection: Specification and Proxy Audit

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Abstract: Reusing a recorded perception corpus on a re-specified sensor rig is attractive for low-speed L4 logistics programmes, but the target platform is usually never driven, so no target-view image exists to score a reprojection against. This article specifies and audits machinery for that regime rather than validating a reconstruction system. A provenance contract partitions every delivered pixel into observed, gate-recovered or generated, locks observed pixels on the rectified internal raster, and forbids visual-only completion from writing depth or pose evidence, with a module permission matrix that makes the partition enforceable in compositing code rather than documented afterwards. An evidence profile - traceable coverage, gate acceptance under an explicit cycle test, generated fraction, locked-pixel compliance, per-camera risk and temporal spread - replaces a photometric score that is undefined without a reference. We audit both on GCR-NVS, a geometry-first seven-camera RGB-LiDAR pipeline: the three classes account for the delivered image exactly and no locked pixel is modified in any of 21 cases. Those are consistency audits, not measurements of fidelity. The demonstration is a proxy - one sequence, parameterized rather than measured target rigs, reprojections retaining the same-name source camera - so no cross-rig synthesis, geometric-accuracy or downstream claim is made.

Keywords: cross-platform camera-data reuse; cross-rig view reprojection; RGB-LiDAR fusion; multi-camera geometry; data provenance; autonomous logistics vehicles; SE(3) rig transformation; visibility reasoning; sensor reconfiguration; auditable evidence profile

1. Introduction

Electric and autonomous vehicle programmes accumulate large volumes of synchronized camera and LiDAR data. Reusing that corpus on a new platform would avoid repeating road collection, recalibration and annotation each time the sensor suite is revised. We take as the premise of this work that low-speed L4 logistics platforms are produced in small volumes and re-specified often enough that a fresh collection campaign per revision is uneconomic. That premise motivates the problem; it is not measured here, and no result in this paper depends on it.

The reuse problem is not a conventional image-translation task. A target vehicle may place its cameras at a different height and baseline, use different focal lengths and distortion, and see surfaces occluded in the source rig, so resizing, cropping or unconstrained generation cannot guarantee a target pixel corresponds to a physically valid scene point. A model trained on silently hallucinated road surface learns a road that was never there.

Neural view synthesis has advanced rapidly, from radiance fields [1-4] to explicit Gaussian representations [5] and feed-forward reconstruction from sparse input [6-8], with diffusion priors where the input is too sparse to constrain the scene [9,10]. Driving reuse demands a property standard image metrics do not express: measured evidence must remain distinguishable from inferred appearance. A hallucinated pixel may look plausible while violating road topology, where a conservative renderer would expose the unknown region instead.

1.1. The No-Ground-Truth Regime

A second difficulty dominates practice. Conventional evaluation needs synchronized imagery from the target vehicle at the same place and time as the source; such a double-rig collection is expensive, must be jointly calibrated, and usually does not exist while the algorithm is being developed.

A common but unsound response improvises a reference from the same-named source camera and reports PSNR or SSIM against it; Section 7.4 shows why that inverts the intended ranking. The correct response is an evidence profile: how much of the target image a calibrated observation supports, how much was recovered under an explicit consistency test, how much was generated, and whether any observed pixel changed.

1.2. Approach

The pipeline pairs a seven-camera RGB-LiDAR source rig with a parameterized seven-view target rig. Two branches, named L4-SE and A-ONLY after the internal components they wrap and carrying no expansion beyond that, share the same calibration, rig transform and provenance contract but report separately: L4-SE supplies calibration, rectification, SE(3) rig motion, LiDAR anchoring and audit artefacts with no learned component; GCR-NVS adds dense learned structure, semantic correspondence and controlled residual completion on the same geometry. Visual polishing never modifies metric depth, LiDAR anchors or rigid-pose optimization.

The design principle can be stated in one sentence: **the pipeline is permitted to make a target image look complete, but it is never permitted to make it look observed.** Everything else in the architecture follows from enforcing that separation at each stage.

1.3. Contributions

Five contributions follow; the first two are the ones we ask to be judged.

1. **A provenance contract.** A per-pixel partition into observed, gate-recovered and generated, with a module permission matrix fixing which stage may write which channel. Observed pixels are locked byte-exactly on the rectified internal raster the lock is defined over; export redistorts by interpolation, so byte equality is claimed before that resampling and not after. No visual-only module may write depth, pose or any reported geometric quantity, and the contract is enforced in the compositing code rather than documented alongside it.
2. **An evaluation protocol for the no-ground-truth regime.** An evidence profile -- observed coverage, gate acceptance under an explicit cycle test, generated fraction, locked-pixel compliance, per-camera risk and temporal spread -- in which every quantity has a defined support and a stated measurement status, replacing a photometric score that is undefined without a reference. Section 5.4 says where the profile is gameable: with the same-name source camera in the input, observed coverage is copy-sensitive and is not a capability measure.

3. **A task taxonomy** separating strict leave-one-camera-out reconstruction, same-name small-baseline reprojection and true cross-vehicle synthesis, and identifying which one each result addresses.
4. **A reference implementation**, geometry-first: metric LiDAR anchors, dense monocular depth candidates, self-supervised semantic correspondence, a rigid SE(3) rig transform and gated residual synthesis under the provenance mask, including the fusion logit that explains why small-baseline results resemble their source images.
5. **A proxy audit**. One sequence, two parameterized rigs, 21 shifted-pose cases and 20 consecutive frames. The three classes account for the delivered image exactly with zero modification of locked pixels. Two findings outlast this pipeline: a seven-camera mean conceals a 4.9-fold concentration of vehicle-region risk in one camera, and a delivered video can be checked against its declared construction from the pixels alone.

What this does not establish: the shifted-pose cases are same-name reprojections, not leave-one-camera-out synthesis; both rigs are parameterized configurations, so no result characterizes a real platform; and we report no geometric error against a depth holdout, no ablation and no downstream retention. Table 7 specifies the eight runs that would close these, seven needing no new collection.

2. Related Work

2.1. View Synthesis, Depth and Visual Features

Neural Radiance Fields render new views by ray integration [1], with extensions to unbounded scenes [3], improved reconstruction [2] and hash-grid primitives [4]; Gaussian Splatting substitutes explicit primitives for real-time rendering [5], and plane-based representations [11,12] trade generality for a cheaper model. All are evaluated by scoring a held-out image from the same capture, which measures interpolation inside the observed view manifold.

At corpus scale per-scene optimization is impractical, so generalizable methods condition on image features [2,6], feed-forward Gaussian methods predict primitives in one pass [7,8], and diffusion approaches treat synthesis as conditional generation [9,10,13]. These are powerful because they hallucinate plausibly, which is why their output is not measured evidence unless the pipeline records which pixels it produced. Ours admits such a prior only behind a mask and marks its output generated.

Dense monocular depth has matured from transformer baselines [14] to large-scale mixed supervision giving robust metric depth [15], but these are a shape prior, not metric ground truth: scale drift, thin-structure smoothing and boundary bleeding are documented failure modes invisible in a depth colourmap. Self-supervised DINOv2 features [16] transfer well to correspondence where photometric consistency is weak, and feature upsampling recovers boundaries patch-token backbones discard [17].

2.2. Calibration, Rigid Rigs and Driving-Data Reuse

We use the Brown-Conrady model [18] with supplied coefficients, compute all geometry in a rectified pinhole domain and remap only for display [19,20], and represent motion in SE(3) [21], within the metric-consistency problem the SLAM literature addresses [22]. A physically meaningful target rig applies one vehicle transform to fixed mounts, whereas independent per-camera warps give images that are individually plausible and jointly impossible (Section 3.3).

Reusing driving data is established: public datasets [23-25] made multi-sensor corpora available, simulators [26] offered configurability at the cost of a domain gap, and sensor-realistic resimulation reconstructs and re-renders the scene [27-35]. Our problem differs in that the target configuration is a supplied file rather than a property of the

collection, and the output feeds perception, which raises an unlabelled hallucination from an aesthetic defect to a data-integrity one.

2.3. Downstream Consumers, Provenance and Positioning

The consumer is usually a perception stack, and modern camera-based stacks lift features into a shared bird's-eye-view space [36,37] and increasingly fuse the whole driving task [38]. Such stacks are sensitive to extrinsics, so their performance on reprojected input follows geometric fidelity rather than perceptual quality, which makes a downstream retention experiment the most informative validation available without target-view ground truth. We do not run one; it is P4 in Table 7.

Separating what a model measured from what it inferred recurs in trustworthy machine learning [39], and image restoration supplies our machinery for residual regions [40–42]. Documentation practice records provenance at corpus and checkpoint level [43,44]; carrying it to the pixel and exporting it is rare, since common practice computes a validity mask internally and discards it. We treat the provenance map as a first-class output alongside RGB and depth (Section 3.4, Table 3): the same kind of accountability record datasheets and model cards established, but per pixel and enforced in code rather than written afterwards.

This work differs from all three lines on four axes at once. Evaluation assumes no target-view reference exists, rather than holding one out from the same capture. Provenance is exported per pixel rather than discarded. Observed pixels are locked, so no later visual module can overwrite evidence. And the target viewpoint is one rigid vehicle transform applied to fixed mounts, not an independent warp per camera. Each has precedent alone; we are not aware of a driving-reuse system enforcing all four together, and it is the combination Section 5 audits.

3. Problem Formulation and Evidence Contract

3.1. Inputs, Outputs and Notation

Let a source key frame contain calibrated RGB images $\{I_i\}$, $i = 1..N$, with intrinsics K_i , Brown-Conrady coefficients D_i , extrinsics T_i and a LiDAR point set $P = \{X_n\}$; in the supplied data $N = 7$. A target camera j is specified by (K_j^t, D_j^t, T_j^t) , read from a rig configuration file rather than hard-coded. The output is not a single image but a bundle:

$$O_j = (\hat{I}_j, \hat{Z}_j, V_j, Q_j, U_j) \quad (1)$$

Extrinsics follow $X_c = T_{c \leftarrow w} X_w$, so the stored matrix maps a world or LiDAR point into the camera frame; the vehicle frame has x forward, y left, z up.

Table 1 summarizes the source rig. Two properties drive the results: the two front cameras have four times the resolution of the surround cameras, and focal lengths are not comparable across the rig. The surrounds sit between 1082 and 1903 px and the front wide at 1922 px, while the front narrow is 7340 px, giving it a 29.3 degree horizontal field of view against 53.5 for the rear camera and 82 to 90 for the other five.

Table 1. Source rig, with focal length and horizontal field of view computed from the supplied calibration. The front narrow camera differs from every other by a factor of at least 3.8 in focal length, which drives the per-camera behaviour of Section 6.4.

Sensor	Native resolution	f_x (px)	Horizontal FOV	Role in the pipeline
CAM_FRONT_WIDE	3840 x 2160	1922	89.9 deg	Forward short-focal; best-supported target camera

Sensor	Native resolution	f_x (px)	Horizontal FOV	Role in the pipeline
CAM_FRONT_NARROW	3840 x 2160	7340	29.3 deg	Forward long-focal; most sensitive to rig displacement
CAM_FRONT_LEFT	1920 x 1080	1085	83.0 deg	Forward-left surround
CAM_FRONT_RIGHT	1920 x 1080	1113	81.5 deg	Forward-right surround
CAM_BACK_LEFT	1920 x 1080	1082	83.2 deg	Rear-left surround
CAM_BACK_RIGHT	1920 x 1080	1087	82.9 deg	Rear-right surround
CAM_BACK	1920 x 1080	1903	53.5 deg	Rear medium-focal
LIDAR_CONCAT	Concatenated sweep	-	-	Metric anchors, occlusion ordering, scale constraint; never rendered as colour

Field of view is $2 \arctan(w / 2f_x)$, the rectified pinhole value rather than the manufacturer's optical specification. Extrinsic array ordering is not lexicographic (Section 4.2).

Figure 1 decodes the same calibration geometrically, recovering each camera centre and optical axis from the per-frame extrinsic. The seven cameras sit 1.38 to 1.43 m above the LiDAR origin and span the full azimuth, but not uniformly: the two forward cameras are nearly co-axial at 2.5 and 2.8 degrees of yaw and differ almost entirely in field of view, while the four surround cameras spread between 46 and 135 degrees. Figure 2 separates the two optical quantities that drive Section 6.4.

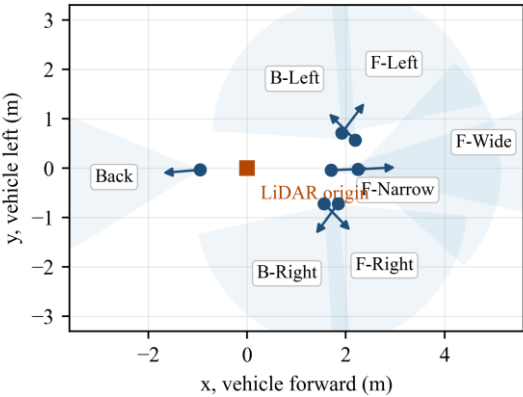


Figure 1. Source rig geometry decoded from the supplied extrinsics. Dots are camera centres, arrows optical axes, wedges the horizontal field of view implied by each focal length; the square is the LiDAR origin. The narrow forward wedge is the camera Section 6.4 identifies as the dominant risk concentration.

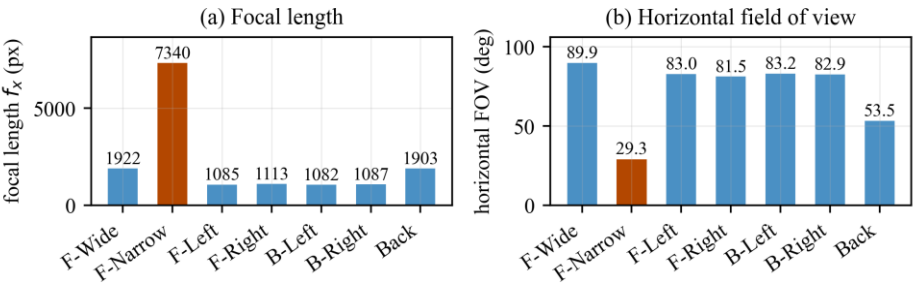


Figure 2. Per-camera optics computed from the calibration. (a) Focal length; (b) horizontal field of view. The front narrow camera, highlighted, is 3.8 times the focal length of the front wide camera and 6.6 to 6.8 times the four surround cameras, which is the geometric cause of its behaviour in Table 12 and Figure 22.

3.2. Three Task Definitions That Must Not Be Conflated

Much of the confusion around cross-vehicle synthesis results comes from reporting one task and implying another. We therefore separate three problems explicitly. Table 2 states them side by side.

Task A, strict leave-one-camera-out reconstruction. Given six of the seven synchronized cameras, camera j is hidden completely and must be reconstructed:

$$\hat{I}_j = F_{\theta}(\{I_i, K_i, D_i, T_i\}_{i \neq j}, P, K_j, D_j, T_j) \quad (2)$$

The true image I_j is used only for supervision and evaluation, never as input. This tests whether the model can exploit neighbouring fields of view, geometry and learned priors to synthesize a direction it cannot see, and it admits an exact photometric ground truth, making it the most trustworthy quantitative proxy available without new collection.

Task B, same-name small-baseline reprojection. Given all seven cameras, each target camera is displaced to a nearby pose:

$$\hat{I}'_j = F_{\theta}(\{I_i, K_i, D_i, T_i\}_{i=1}^7, P, K'_j, D'_j, T'_j) \quad (3)$$

Because the input still contains I_j at its original pose and a translation-only perturbation leaves the optical axis unchanged, the same-name source camera overlaps the target field of view almost completely. This validates that the target pose enters the projection equations, that extrinsic directions are correct and that motion is continuous; it does not on its own demonstrate novel-view synthesis capability.

Task C, true cross-vehicle target synthesis. The target cameras may differ in position, orientation, intrinsics and field of view:

$$\hat{I}_j^{L^4} = F_{\theta}(\{I_i^{src}, K_i^{src}, D_i^{src}, T_i^{src}\}, P, K_j^{L^4}, D_j^{L^4}, T_j^{L^4}) \quad (4)$$

Throughout, j indexes a camera and a superscript t marks a target-side quantity, so K_j^t , D_j^t and T_j^t denote the general target camera j of Table 5 and the task-specific forms above are instantiations of it. The two roles are deliberately kept on different levels: writing the held-out camera as a subscript t as well would make the same symbol an index in one equation and a label in the next. Indices are written explicitly because the pipeline manipulates three simultaneous index families -- source camera, target camera and depth hypothesis -- and collapsing them would hide which one a sum runs over. If the target platform was absent at collection time no pixel-level ground truth exists for Task C, which can then only be validated by a double-rig collection, a jointly calibrated pair of platforms, leave-one-out on real target cameras, or indirectly through geometric, temporal and downstream consistency.

Task C is the operational goal, Task A the most credible quantifiable proxy, Task B a geometric stress test. **The experiments reported here are Task B plus a geometric audit of a Task C configuration; they are not Task A.** The numbers in Section 6 would mislead if read as leave-one-out results, and the same-name camera in the input is the principal reason the reconstructions resemble the source images.

Table 2. Three task definitions frequently conflated in cross-rig reporting. The experiments here are Task B plus a geometric audit of a Task C configuration; no Task A result is reported.

Task	Definition	Ground truth available	What a good result proves
A. Strict leave-one-out	Hide camera j entirely; reconstruct it from the other six plus LiDAR	Yes, the hidden image itself	The model exploits neighbouring fields of view and geometry to synthesize an unseen direction

Task	Definition	Ground truth available	What a good result proves
B. Same-name small-baseline	All seven cameras given; each target camera displaced to a nearby pose	No, but the same-name source view is nearly aligned	The target pose enters the projection correctly; extrinsics, depth and continuity are consistent
C. True cross-vehicle	Target cameras differ in position, orientation, intrinsics and field of view	No, unless a synchronized target platform was driven	The pipeline transfers a corpus to a new sensor configuration

Reporting a Task B number as though it answered Task A is the specific error this table exists to prevent.

3.3. The Rigid Target Rig

The target rig is parameterized by a single shared vehicle-to-rig pose and a set of fixed camera mounts:

$$T_{L \leftarrow C_j}^{\text{view}} = T_{L \leftarrow R}^{\text{view}} \cdot T_{R \leftarrow C_j}^{\text{mount}} \quad (5)$$

This factorization is the constraint that keeps the seven target images mutually consistent: one degree-of-freedom set moves the whole module and the mount matrices fix the relative arrangement. Seven independent warps would instead give images that are individually plausible but jointly correspond to no physical sensor arrangement, which no per-camera metric detects.

When a target camera centre is displaced by ΔC in the vehicle frame with the rotation held fixed, the stored extrinsic translation updates as

$$t' = t - R\Delta C \quad (6)$$

and the perturbation is accumulated onto the base rig offset, $C_j' = C_j^{\text{base}} + \Delta C$, rather than overwriting it. This preserves the base rig geometry under a stress test, which is what makes the three perturbation bands of Section 5.3 comparable to one another.

3.4. The Provenance Contract and Its Cost

Each output pixel is assigned to exactly one of three evidence classes:

$$Q(p) \in \{\text{observed, gate, generated}\} \quad (7)$$

Observed pixels are sampled from a calibrated source image through a geometrically valid correspondence and are then locked: no downstream module may alter their value. The label asserts source-image ancestry under a correspondence that is itself estimated, not that the target surface was measured: the depth carrying the correspondence is a dense candidate, so a pixel can be observed in this sense and still land on the wrong surface. Only the LiDAR-anchor subtype below is a direct range measurement. We keep the name because the exported masks and the released code use it, and read *source-supported* whenever the stronger reading would matter. The lock is defined on the rectified internal raster, where every geometric export and the locked-pixel count of Section 6.5 are computed. The redistortion of Section 4.12 resamples that raster into the target camera's distortion domain, so exported display pixels are interpolated by construction: the guarantee is that no module changed an observed value beforehand. **Gate** pixels are geometric holes filled only after a source-target-source cycle and the visibility checks of Section 4.10 gave positive supporting evidence. **Generated** pixels occupy the residual holes, are visual-only and are excluded from every metric and geometric export. The masks are written to disk alongside the RGB, and the system never treats generated RGB as depth or pose evidence.

Two further states are not labels of Equation (7). A *LiDAR anchor* is a subtype of observed, restored byte-exactly after the display polish. *Unknown* is the complement of the partition; the delivered RGB is hole-free by construction, so it is empty in the image and

survives only as an exported mask. Only the three labels sum to the evidence budget of Table 13.

The contract is a permission matrix over modules, not a convention. Table 3 records which module may write which channel; two rules carry the weight. No learned module may write into the observed mask, and no module whose output is generated may contribute to the depth export, the SE(3) optimization or any reported geometric metric.

Table 3. The provenance partition and module permission matrix. The upper block is the three-way partition of Equation (7); the lower block lists two support states that are not labels of it.

Class	Admission condition	Exports depth	SE(3) eligible	May be modified later by
Observed	Valid geometric correspondence to a calibrated source pixel	Yes	Yes	Nothing; excluded from the modification mask of Equation (20) and from the display polish, so locked byte-exactly
Gate	Hole pixel passing the source-target-source cycle, depth-ordering and dynamic-region tests	Yes	Conditional	Residual decoder of Equation (20) wherever the geometric budget is non-zero, and display polish; a gate pixel the decoder writes is relabelled generated and loses depth eligibility
Generated	Residual hole filled by the masked completion module	No	No	Display polish only
LiDAR anchor (support state)	Direct LiDAR return surviving the z-buffer; a subtype of observed	Yes	Yes	Nothing; locked as observed, and additionally restored byte-exactly after the polish as a redundant safeguard
Unknown (support state)	Hole rejected by the gate and not completed; empty in the delivered RGB	No	No	Exported as a mask; never written into RGB as evidence

The contract costs something real: the delivered geometry declines to reconstruct regions a vendor optimizing for demonstration quality would fill, and those regions grow with baseline. What it buys is that every quantitative statement has a defined support. When Section 6 reports 98.24% observed coverage, that refers to pixels each traceable to a specific source camera, pixel coordinate and calibration.

4. Materials and Methods

4.1. Pipeline Overview

Figure 3 shows the end-to-end architecture. Processing proceeds in five stages: rectification and metric anchoring, dense structure estimation, target rendering with visibility reasoning, gated recovery and residual completion, and export with redistortion. Geometric stages complete before any visual-only stage, so a geometric result stays reproducible when the optional visual models are unavailable.

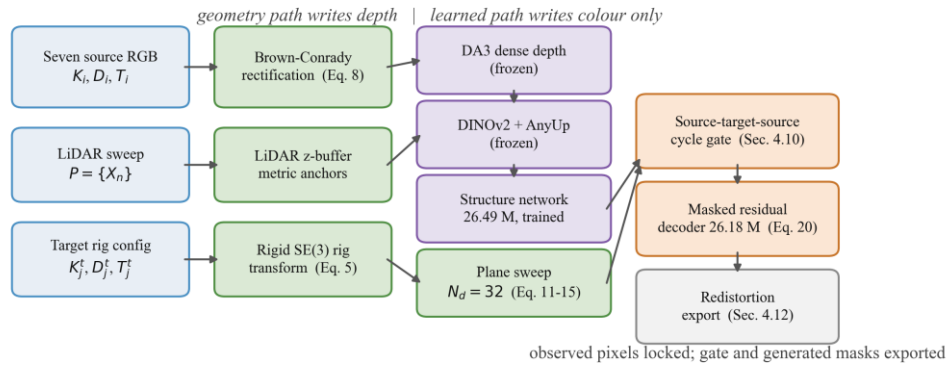


Figure 3. End-to-end architecture.

4.2. Rectification, the Intrinsic Domain and a Calibration-Ordering Audit

Raw images are rectified with the calibrated Brown-Conrady coefficients, producing an internal pinhole domain with intrinsic matrix K'_i and zero distortion:

$$I^{\text{rect}} = \text{Remap}(I^{\text{raw}}, K, D \rightarrow K', 0) \quad (8)$$

All geometric computation is performed in this rectified domain; only at export is the result remapped to the target camera's physical distortion domain. Mixing the two is a silent error: a distorted-domain projection is subtly wrong in the periphery in a way that reads as mild miscalibration rather than a bug.

For a three-dimensional point X in the source frame, projection is

$$\tilde{p}_i = K'_i T_i X, p_i = (\tilde{p}_{i,x}/\tilde{p}_{i,z}, \tilde{p}_{i,y}/\tilde{p}_{i,z}) \quad (9)$$

and a point is valid only when its camera-frame depth is positive and its projection falls inside the image bounds.

One implementation finding is invisible to unit tests and expensive to diagnose. The per-frame calibration stores the seven extrinsics in a non-lexicographic order: back-left, front-left, front-right, back-right, back, front-wide, front-narrow. Consuming it in name order pairs each image with a neighbouring camera's extrinsic and yields reprojections that are self-consistent but physically wrong, with a half-frame void, mirrored parallax and anomalously low coverage.

4.3. LiDAR Anchoring and Visibility Rasterization

LiDAR points are transformed into each rectified source camera and rasterized with a z-buffer, fixing the metric scale, establishing occlusion ordering and constituting the hard evidence that constrains learned structure; anchor pixels survive every later visual stage byte-exactly. LiDAR is deliberately not used to paint colour: a concatenated sweep is far too sparse for sky, distant facades and many object surfaces, and rendering points as RGB gives an image whose apparent density reflects point density rather than scene content.

4.4. The Geometry-Only Branch and Source Precedence

The L4-SE branch composes the target image from the geometric sources of Table 14 in strict priority order, each writing only where every higher-priority source has declined: LiDAR anchors; the current rectified RGB-D surface; temporally adjacent frames of the same camera, after an SE(3) warp and the consistency gates; adjacent calibrated cameras, under rigid-rig reprojection and visibility tests; and a sparse LiDAR-anchored TSDF. These five are the only classes that may write depth. Everything below them in Table 14 is visual-only and excluded from every geometric export, which is why the branch runs without any learned component and why its coverage in Tables 8 and 10 is re-derivable from calibration and point cloud alone.

4.5. Dense Structure Candidates and the Structure-Constraint Network

The learned branch predicts a dense metric depth candidate d_{DA3} from the multi-camera RGB input using a large metric variant of the Depth Anything family [15]. It is dense and well aligned with image boundaries, but its metric scale is not reliable at the accuracy the downstream reprojection needs. A structure-constraint network corrects local scale and surface residual using LiDAR support, temporal registration, confidence, dynamic-region masks and RGB boundaries:

$$d^*(p) = d_{DA3}(p) \cdot \exp(g(p) \cdot r_\theta(p)) \quad (10)$$

where r_θ is the predicted log-residual and $g(p) \in [0,1]$ is a bounded gate driven by the distance to the nearest LiDAR support, the local confidence, the dynamic-object mask and the RGB boundary map. The multiplicative log-residual parameterization is chosen so that the correction is scale-relative and so that a zero residual reproduces the candidate exactly.

Two constraints are enforced, not encouraged: the network may not overwrite a LiDAR anchor, and it writes depth only, never colour. Note that d^* is an estimate whose accuracy must be measured against an independent holdout, since a self-consistency check on the constraining points would report a small error regardless of whether the surface is right.

4.6. Learned Module Specification

Table 4 specifies every learned module and where the trained capacity sits. Three points matter. The trained capacity is small and concentrated: 52.76 M parameters, the structure network holding 26.49 M and the residual decoder 26.18 M, against three frozen foundation models used as published. The structure network is a depth model, not an appearance model: 48 input channels, 16 geometric and 32 semantic, and four outputs, none of them colour. And its log-residual is bounded at 0.22, so one forward pass rescales a depth by at most about 25% — it corrects the dense candidate rather than replacing it.

Table 4. Learned modules of the released inference configuration. Parameter counts are obtained by instantiating the released code, not estimated. Frozen modules are used as published and are not fine-tuned on this data.

Module	Architecture	Role and constraint	Params	State
Dense depth (DA3Metric-Large)	External ViT	Dense metric depth candidate; not ground truth, scale set by LiDAR	-	Frozen
Semantic encoder (DINOv2)	ViT-L/14 at 364 x 644; ViT-B/14 conditions the structure net	Semantic context and correspondence; writes no colour, no depth	-	Frozen
Feature upsampler (AnyUp)	Multi-backbone head	Restores high-resolution feature boundaries	-	Frozen
Structure network	4-stage U-Net, 48 base channels, 384-channel bottleneck, 3-layer global transformer	Corrects DA3 scale and surface residual under LiDAR; writes depth only, never colour, never overwrites an anchor	26.49 M	Trained
Residual decoder	Restormer-Small, dim 48, blocks (4, 6, 6, 8)	Predicts RGB residual, modification strength and log uncertainty; restricted to the mask of Equation (20)	26.18 M	Trained
Feature adapter	48 -> 32 channel Restormer adapter	Projects fused features for the decoder	0.09 M	Trained
Trained total			52.76 M	

The structure network takes 48 input channels, 16 geometric and 32 semantic, and emits four: log-residual, correction gate, confidence and depth-boundary probability. Its log-residual is bounded at 0.22, so a single forward pass can rescale a depth by at most about 25%. The released checkpoints were trained at 320 × 180 with 48 base channels and are frozen for every result in this paper.

4.7. Semantic and Detail Encoding with Anchor Attention

Frozen DINOv2 features [16] give semantic context and are stable under the illumination changes that break photometric matching; the released backbone is a frozen ViT-L/14 of hidden dimension 1024, with ImageNet-normalized inputs at 364 × 644, while a ViT-B/14 of dimension 768 conditions the structure network and patch tokens from several intermediate layers. Freezing is deliberate: the training corpus is small relative to the encoder's capacity.

Because a patch-token backbone discards the high-frequency structure that matters for lane markings, poles and barriers, a lightweight convolutional detail branch of NAFNet-style blocks [42] runs in parallel at higher resolution and feature upsampling [17] restores boundary sharpness before fusion.

For each selected LiDAR anchor X_n , features are sampled from every source camera in which it is visible and aggregated with a four-head attention block [45], so the aggregation learns which views to trust rather than averaging them. The rendered anchor feature map supplements dense geometry and may not alter anchor depth.

4.8. Plane-Sweep Stereo and Back-Projection

The target depth and a coarse target RGB are estimated by a plane sweep in the target view, following the classical space-sweep formulation [46] and its learned cost-volume descendants [47]. Candidate depths are sampled uniformly in inverse depth over the configured metric range,

$$q_k = \frac{1}{d_{max}} + \frac{k}{N_d - 1} \left(\frac{1}{d_{min}} - \frac{1}{d_{max}} \right), d_k = \frac{1}{q_k} \quad (11)$$

with $N_d = 32$ layers over a default range of 2 to 80 m. Inverse-depth sampling places the layers more densely at short range, matching the sensitivity of perspective displacement to depth.

Each source image is encoded by a three-layer convolutional stem to a quarter-resolution 32-channel feature map F_i , and for each candidate depth the source features are warped into the target view. A variance cost is accumulated over the valid views,

$$\mu_k(p) = \frac{1}{N_k} \sum_i M_{i,k}(p) F_{i,k}(p) \quad (12)$$

$$V_k(p) = \frac{1}{N_k} \sum_i M_{i,k}(p) F_{i,k}^2(p) - \mu_k^2(p) \quad (13)$$

and locations supported by fewer than two views receive an explicit penalty, so that single-view coverage is not mistaken for multi-view agreement. This penalty matters more than it appears: without it, the periphery of the field of view, where only one camera sees the scene, produces confidently wrong depth.

$$P_k(p) = \frac{\exp(-c_k(p))}{\sum_j \exp(-c_j(p))}, \hat{Z}(p) = \sum_k P_k(p) d_k \quad (14)$$

and a normalized-entropy confidence is derived from the same distribution,

$$C_d(p) = 1 - \frac{-\sum_k P_k(p) \log P_k(p)}{\log N_d} \quad (15)$$

A peaked distribution yields high confidence and an ambiguous one yields low confidence. This confidence is subsequently consumed by the modification mask of Equation (20) in Section 4.11, which is how uncertainty in depth is translated into permission to repaint.

For a pixel p_j of target camera j at candidate depth d , the target ray is back-projected, transformed to the world frame and then into each source camera:

$$X_j(d) = d \cdot (K_j^t)^{-1} \bar{p}_j, X_w(d) = (T_j^t)^{-1} X_j(d), X_i(d) = T_{i \leftarrow w} X_w(d) \quad (16)$$

and finally projected as $p_i(d) = \pi(K_i X_i(d))$. Bilinear sampling then yields the source colour or feature. This single chain is the shared geometric basis of the plane sweep, the full-resolution RGB warp and anchor feature sampling; implementing it once and reusing it is what keeps the three consistent.

LiDAR restricts the global search range before the sweep,

$$d'_{min} = \max(0.5 \cdot \min Z_L, d_{min}), d'_{max} = \min(1.5 \cdot \max Z_L, d_{max}) \quad (17)$$

and afterwards a local correction compares the swept depth with LiDAR depth in a 15×15 neighbourhood, with the correction magnitude clamped to 25% of the current prediction. Where LiDAR support is strong the confidence floor is raised to approximately 0.8. The clamp is important: an unclamped correction lets a single spurious return drag a whole neighbourhood, and the resulting artefact is a locally plausible surface at the wrong distance, which is exactly the class of error the provenance contract is meant to exclude.

4.9. Full-Resolution Warping and Multi-View Fusion

Depth is estimated at quarter resolution but colour is warped at full resolution using that depth, so high-frequency texture comes from the source sensor rather than an upsampled intermediate. Each source image is warped once into the target view, giving $I_{\sim i}$ and a validity mask M_i , and the views are fused by a softmax over per-view logits:

$$w_i(p) = \frac{M_i(p) e^{\ell_i(p)}}{\sum_r M_r(p) e^{\ell_r(p)}}, I_c(p) = \sum_i w_i(p) \tilde{I}_i(p) \quad (18)$$

with the logit composed of a reprojection-error term, a feathering weight, an optical-axis similarity term, a baseline penalty and a same-direction bonus:

$$\ell_i = -\frac{e_i}{0.05} + \log f_i + 2 \log s_i - \frac{b_i}{0.50} + 16 \cdot \mathbf{1}(s_i > 0.9999 \wedge b_i < 0.50) \quad (19)$$

Here e_i is the residual against the multi-view consensus, f_i a feathering weight near the valid-region boundary, s_i the cosine similarity of source and target optical axes, and b_i the baseline in metres; the fused weights are smoothed with a 15×15 kernel to suppress seams.

The final term is the single most important caveat on Section 6. A source camera aligned with the target within tolerance and within 0.50 m of it gains an additive 16 on its logit, which dominates the softmax over the configured range of the other terms. In Task B the same-name camera is nearest by construction and is intended to satisfy both conditions, but we cannot show it does so in every case: the realised baselines are not exported, the widest band reaches 0.50 m alone, and any base-rig offset adds to it, so boundary cases may lose the bonus. Reporting the realised baseline and whether the bonus fired is protocol P2 of Table 7. The reconstruction there is the same-name source image plus estimated depth: a small-baseline reprojection with a modest neural correction.

4.10. The Source-Target-Source Cycle Gate

Geometric holes are target regions with no valid warped source support. Some are recoverable, the surface having been observed while the correspondence failed on a depth-ordering ambiguity, a boundary effect or a marginal validity test; the rest are genuinely unobserved. The gate separates them by requiring a closed cycle: a candidate is projected into a source observation and back, and accepted only if the position, depth and colour residuals fall within bounds, the depth ordering agrees with the target z-buffer, and the region is not flagged dynamic.

These values are fixed in the released inference configuration. Depth ordering accepts a candidate whose camera-space depth exceeds the z-buffer by at most $\max(0.10 \text{ m}, 0.02 z)$, so thin distant structure is not culled by depth noise. The dynamic-region test is a Mask R-CNN ResNet-50-FPN segmenter [48] at score threshold 0.5 over person, bicycle, car, motorcycle, bus and truck, dilated by 5 pixels. Fusion uses the temperatures of Equation (19), 0.05 photometric and 0.50 baseline, and drops views below a 0.1 cosine-similarity floor. Equation (18) normalizes over the valid views only, so an invalid view removes a contributor without darkening the pixel, and the plane-sweep path re-masks and renormalizes after smoothing; where no view is valid the pixel passes to the gate as a hole. Failed candidates remain unknown, which is why gate fill falls as the perturbation grows.

4.11. Controlled Residual Completion

The coarse render and geometry channels are concatenated into a 48-channel tensor, 13 of target geometry, 3 of target ray direction and 32 of anchor features, and processed by a Restormer-Small decoder [41] predicting an RGB residual R , a modification strength α and a log uncertainty s . The uncertainty is exported as a diagnostic only: it does not enter Equation (20) or the composition, is not calibrated, and no claim depends on it.

$$M_{\text{geo}} = \text{clip}((1 - G(p)) + G(p)(1 - \kappa(p)), 0, 1), M = M_{\text{geo}} \cdot \mathbf{1}[Q(p) \neq \text{observed}], \alpha_{\text{eff}} = \alpha \cdot M \quad (20)$$

where $G(p)$ in $[0, 1]$ indicates that target geometry exists at p and $\kappa(p)$ in $[0, 1]$ is its confidence, so the decoder may act where geometry is absent or uncertain **and** the pixel is not labelled observed. Without the indicator an observed pixel with confidence below one would receive a non-zero budget; with it, the locked-pixel result of Section 6.5 is a property of Equation (20) rather than an observation. The final internal-domain image is composed as

$$\hat{I} = (1 - \alpha_{\text{eff}})I_c + \alpha_{\text{eff}} \cdot \text{clip}(I_c + R, 0, 1) \quad (21)$$

Three transitions follow, since a partition without transition rules is underspecified the moment a stage writes twice. Observed admits no transition: the indicator removes it from the modification mask, which is what Section 6.5 measures. A gate pixel the residual decoder writes becomes generated and loses its depth eligibility, because a visual-only module may not support a geometric export. Generated does not transition further. The rule is conservative by construction: where a label is in doubt the lower-evidence one wins, so the observed class can only shrink under processing.

An optional adapter conditioned on the masked RGB, depth, semantic features and hole mask is applied only to residual holes surviving the gate; we call it the **A-ONLY adapter**, which the artefact registry names *residual adapter*. A large-baseline video-diffusion prior may attach here, strictly as a completion component and never as geometric evidence. Whatever these modules write is labelled generated, excluded from depth and SE(3) optimization, and remapped to the display domain only after every geometric export is committed.

4.12. Export, Reproducibility and Cost

For every camera and frame the pipeline exports RGB, depth, validity, provenance, hole and unknown masks, a machine-readable report and a lineage record covering source sequence and frame, target rig, resolution, code revision, checkpoint hash, seed, device, elapsed time and the metric scope under which the numbers may be quoted — the field that stops a geometry-branch figure being quoted later as a learned-branch result. Render times of roughly 8 to 31 seconds per camera make this an offline conversion workload, not a real-time renderer.

5. Experimental Protocol

5.1. Data

The principal development sequence, 2026-05-22-10-25-21, contains 80 synchronized key frames spanning about 39 seconds, seven RGB cameras and one concatenated LiDAR stream, with per-frame calibration supplying the extrinsics, intrinsics and image dimensions and a separate file the distortion coefficients.

Frame 40 serves the geometry acceptance audit, frame 73 the learned-branch shifted-pose evaluation, and frames 24 to 43 the temporal series. The data are challenge reference data under a use agreement and are not redistributed; Section 8 and the Data Availability Statement describe lineage and access.

The supplied calibration omits several quantities a fully specified cross-rig transform needs: the LiDAR origin height, the absolute camera mounting positions in a body frame, the body-to-LiDAR extrinsic and the source vehicle dimensions. Where needed the pipeline uses documented team estimates, marked as assumptions in the configuration files.

The LiDAR stream that anchors every geometry decision also bounds what the observed class can contain. Figure 4 renders the concatenated sweep top-down under both rig definitions and Figure 5 summarizes its geometry. The target-frame cloud holds 313,196 returns with a median planar range of 11.45 m; 86.5% lie inside 30 m and 93.5% inside 50 m, the 95th percentile reaches 54.57 m and the farthest return 250.13 m. Half the geometric evidence supporting any synthesized view therefore comes from within about 11 m.

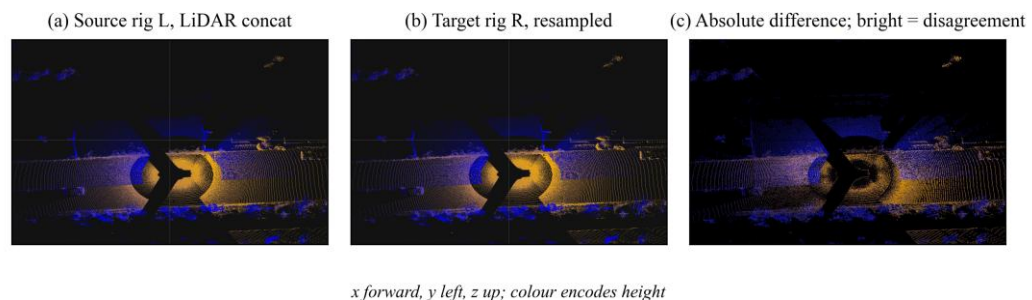


Figure 4. Top-down view of the concatenated LiDAR under both rig definitions. (a) The source rig L; (b) the same sweep in the target rig R; (c) the absolute difference, bright pixels marking cells the two rigs disagree about. Colour encodes height throughout. The dark wedges behind the platform are the near-field sectors of Figure 5(b), a property of the capture vehicle rather than the reconstruction.

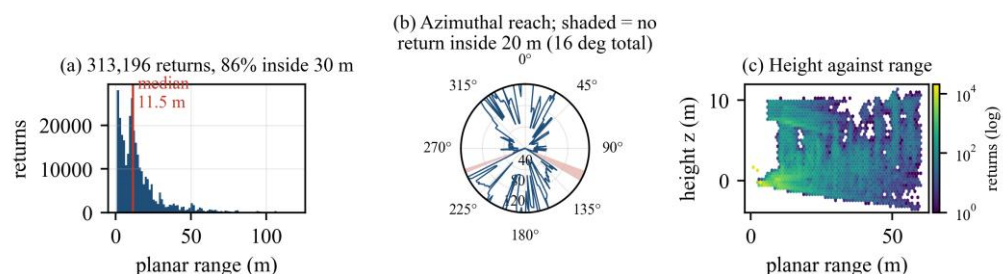


Figure 5. Geometry of the delivered target-frame cloud, 313,196 returns. (a) Planar range, strongly near-field, median 11.45 m; (b) azimuthal reach, shaded over the 16 degrees of azimuth carrying no return inside 20 m; (c) height against range, with the ground plane near $z = 0$ and the sensor's ring structure. Panel (a) bounds what the observed class can contain: 86.5% of returns lie within about 30 m.

Two azimuthal sectors carry no return inside 20 m: one of 11 degrees at 110 to 121 degrees from the forward axis, one of 5 degrees at -113 to -108 degrees. Their near-mirror symmetry about the longitudinal axis is what self-occlusion by the ego platform would produce, not a property of the scene. These are the dark wedges in Figure 4, and any target-rig ray falling inside them is unobservable at close range whatever the pose — a floor on unknown coverage for any method relying on this sweep for geometric support; LiDAR silence does not rule out RGB or temporal evidence, and we do not test whether those could recover the sectors.

The top-down comparison also quantifies rig overlap. Of the cells occupied under either rig, 68.0% are occupied under both, with 16.0% exclusive to each — consistent with a rigid transform, which relocates structure without creating or destroying it; an asymmetry would have indicated an error in the transform. Panel (c) of Figure 4 was verified to be an absolute difference rather than assumed: it correlates at 0.85 with the absolute difference of the inputs against 0.65 with their mean.

5.2. Two Target Rig Configurations

Two target configurations are kept distinct throughout; Table 5 records the parameters that distinguish them; complete intrinsics, distortion coefficients and mount transforms live in the configuration files it names. Rig R1 is the open seven-view reference configuration on a $3.70 \times 1.40 \times 2.00$ m low-speed logistics preset and expresses the intended cross-platform scenario; rig R2 applies a controlled upward translation of about 0.20 m while retaining the source camera arrangement, isolating a pure vertical displacement.

Table 5. The two parameterized target rig configurations. Neither is a measured target-vehicle calibration; both are configuration files read at run time.

Property	Rig R1: open reference preset	Rig R2: audit rig
Configuration identifier	target_l4_rig_reference_v1	20260818 rig audit configuration
Intended role	Expresses the cross-platform scenario	Audit configuration; differs from R1 in height, forward translation, pitch and delivery raster together, so it isolates no single degree of freedom
Vehicle preset	$3.70 \times 1.40 \times 2.00$ m low-speed logistics reference	Source relative arrangement retained
Rig motion vs. source	0.10 m forward, 0.35 m up, 2 degrees pitch	Approximately 0.20 m up
Camera count	7, moved as one rigid module	7, moved as one rigid module
Output resolution	960×540 delivery, audited at source-native raster	1920×1080
LiDAR ground height	0.50 m, team estimate	0.50 m, the same estimate as R1
Target GT available	No	No

Both apply a single vehicle-to-rig transform to fixed camera mounts per Equation (5). Results under the two rigs are reported separately and must not be pooled.

Neither rig is a measured target-vehicle calibration. Both are configuration files: R1 a reference configuration on an open vehicle preset, R2 an audit configuration isolating one degree of freedom. Obtaining a real calibration would mean replacing the configuration and regenerating results, and would invalidate any mixing of old and new numbers, so every report records its rig identifier and Section 6 reports per rig rather than pooled.

The consequence for interpretation is worth stating before any number is read. **No result in this paper characterizes a real target vehicle.** The rigs are instruments for exercising the pipeline and protocol under controlled, known displacements, so Section 7.6's statements about fleet practice are hypotheses to test rather than validated platform findings.

5.3. Evaluation Runs

Three families of runs are reported. **Geometry acceptance:** the geometry branch at frame 40, seven cameras, both rigs, giving Table 8 and a separate hole-free check. **Temporal series:** the geometry branch on frames 24 to 43 for one camera under R1, no temporal smoothing, giving Table 10. **Shifted-pose stress test:** the learned A-ONLY route at frame 73 for 21 cases, seven cameras under each of three bands of 5-10, 10-20 and 20-50 cm, perturbations accumulated per Equation (6). The closest source camera is in the input, so this family is Task B.

The released inference branch uses frozen structure and A-ONLY checkpoints; no training happens during evaluation. Two resolutions must not be conflated: the shifted-pose audit under R2 is rasterized at 1920×1080 , while R1 declares a 960×540 delivery resolution whose geometry audits are computed on the source-native raster and so are unaffected. The audit uses a 32-layer inverse-depth sweep on a single RTX 3090 with 24 GB.

5.4. Metrics

Let Ω denote the target image domain, A the set of pixels with traceable observed RGB support, and H the geometric hole set. We report four primary quantities:

$$\text{ObservedCoverage} = \frac{|A|}{|\Omega|}, \text{GateFill} = \frac{|H_{\text{gate}}|}{|H|} \quad (22)$$

$$\text{GeneratedFraction} = \frac{|H_{\text{generated}}|}{|\Omega|}, \text{LockedChange} = |\{p \in A: \hat{I}(p) \neq I_{\text{locked}}(p)\}| \quad (23)$$

Two denominators are easy to misread. GateFill is a fraction of the hole, not the image: a 44.90% gate fill does not mean 44.90% of the image was recovered. GeneratedFraction is a fraction of the image, bounding the generative module's total influence. LockedChange is an absolute pixel count whose target is exactly zero. We deliberately prescribe no universal accept or reject thresholds for the profile: the protocol reports evidence, and what counts as an acceptable profile depends on the rig, the scene and the downstream use.

One property bounds every conclusion drawn from the profile: **no component of it measures correctness**. Observed coverage counts pixels with traceable source support, not pixels whose colour is right. Gate fill counts candidates surviving a cycle test computed on the pipeline's own estimated geometry, so it measures internal consistency, not agreement with the world. The locked-pixel count checks a constraint the implementation enforces structurally. The profile is a traceability and self-consistency instrument; correctness needs an external reference, which protocols P1 to P3 of Table 7 would supply and this evidence base does not contain.

We also report the vehicle-risk fraction inside holes, computed against the pipeline's own vehicle-region mask, which monitors whether the completion module is being asked to invent traffic participants; a team reusing this diagnostic needs an equivalent mask of its own, and the final visual coverage and hole-free pass rate, which describe the delivered display image rather than the geometry.

Table 6 lists the metric set with its measurement status, which is as much part of the result as the values. Geometry error, semantic preservation and temporal anomaly rate are unmeasured with no proxy substituted; target-view PSNR, SSIM and LPIPS are not applicable rather than unmeasured, because here they would be undefined or compare against the wrong image (Section 7.4).

Table 6. Metric set with measurement status. The status column is part of the result: a quantity marked unmeasured has no substitute reported in its place.

Metric	Definition and denominator	Status
Camera success rate	Fraction of requested target cameras that produced a complete output bundle	Measured
Geometry coverage	Pixels with trusted structural support, over the full frame	Measured
Unknown fraction	Pixels with neither observed support nor gate acceptance, measured on the geometry export before any completion, over the full frame	Measured
Observed coverage	Pixels with traceable calibrated source RGB support, over the full frame	Measured
Gate fill of hole	Hole pixels accepted by the cycle gate, over the hole set	Measured
Generated fraction	Pixels written by the completion module, over the full frame	Measured proxy
Locked-pixel change	Count of observed pixels whose value changed; target is exactly zero	Measured
Vehicle risk in hole	Hole pixels intersecting vehicle regions, over the hole set	Measured
Final visual coverage	Pixels with a delivered colour, over the full frame; includes visual-only fill	Measured
Near-black artefact audit	Suspect pixels and largest connected component among replaced pixels	Measured
Coverage temporal spread	Standard deviation of geometry coverage over consecutive frames	Measured
Geometric error, semantic preservation, temporal anomaly rate	Depth MAE/RMSE against held-out LiDAR; feature similarity or mIoU on observed and gated pixels; optical-flow warping error and track continuity	Unmeasured
Artefact rate	Tears, ghosts and anomalous edges as connected components on the difference map	Proxy only
Downstream retention	Detection and segmentation performance on reprojected versus original data	Unmeasured
Target-view PSNR / SSIM / LPIPS	Photometric fidelity against a target-view reference	Not applicable

Not applicable differs from unmeasured: the three photometric scores are undefined without a target-view reference, and improvising one from the same-named source camera would reward copying the input (Section 7.4).

5.5. What Is Deliberately Not Reported

Four absences are intentional and each is a real limitation. No target-view photometric score is reported, for the reason above. No component-wise ablation is reported: the package contains no matched-seed runs against a common target, and estimating gains from the available artefacts would be fabrication. No baseline comparison is reported, because none has been run on this data under this protocol. And no geometric-accuracy figure is reported, because the depth field is LiDAR-constrained with no subset held out.

All four are closable without new collection. Table 7 specifies the runs: P5 the component ablation, P6 the same-protocol baseline, P3 the LiDAR holdout for the missing geometric accuracy, and P1 and P2 a setting in which a photometric score is defined rather than improvised. Executing P1 to P6 is the necessary condition for any claim beyond the protocol contribution itself.

Table 7. The runs required to convert the present proxy evidence into a benchmark, in ascending order of collection cost.

Protocol	Procedure	New collection needed	What it would establish
P1. Strict leave-one-out	Hide one camera entirely; reconstruct from the other six plus LiDAR; use the hidden image only for evaluation	None	Task A capability with exact photometric ground truth, making PSNR, SSIM and LPIPS meaningful
P2. Nearest-source removal	Repeat the shifted-pose protocol with the nearest source camera removed from the input	None	Removes the same-name shortcut of Equation (19); isolates genuine cross-camera inference
P3. LiDAR holdout	Withhold a LiDAR subset from all constraints; evaluate depth against it in the target frame	None	Geometric accuracy: depth MAE and RMSE, boundary error, near / mid / far buckets
P4. Downstream retention	Run an unmodified perception stack on reprojected and original data and compare	None	Fitness for purpose, the criterion that matters most in production
P5. Component ablation	Disable the depth candidate, semantic features, anchor attention and residual decoder in turn, matched seeds, common target	None	The marginal contribution of each learned module
P6. Same-protocol baseline	Run a published reconstruction or synthesis system on this sequence, rig and metric set	None	Whether the evidence profile beats an off-the-shelf alternative on identical data
P8. Export-domain compliance	Export the rectified and redistorted rasters with the provenance mask; compare the delivered image against the locked reference carried through the same remap, and count cross-class contamination at class boundaries	None	Whether the byte-exact lock of Section 3.4 survives redistortion, and how much of any change is interpolation at class boundaries
P7. Synchronized double rig	Drive both platforms together with joint calibration and measured target poses	Yes	A true Task C benchmark with independent RGB and depth ground truth

6. Results

Every result below is a proxy measurement under the constraints of Section 5.5: the geometry results audit two parameterized rigs rather than a vehicle, and the learned-branch results are Task B (Section 3.2). None is a fidelity measurement against an independent reference, and none should be cited as evidence of synthesis capability or perception-readiness.

6.1. Geometry Acceptance under Two Target Rigs

Table 8 reports the seven-camera geometry audit at frame 40 under both rigs; all fourteen camera requests completed, and Figure 6 plots the per-camera values behind both means. Under R1, the configuration expressing the intended cross-platform displacement, mean geometry coverage is 94.07% with 5.94% unknown; under R2, the pure 0.20 m vertical audit displacement, it is 96.66% with 3.34% unknown.

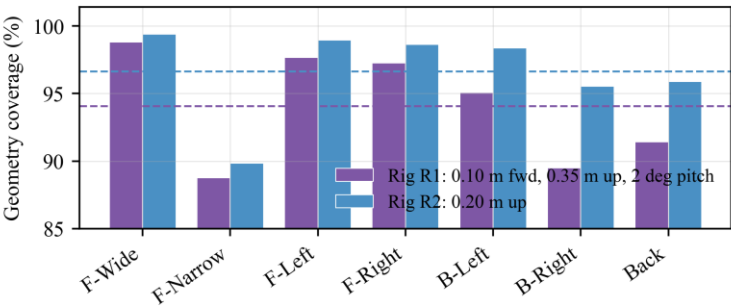


Figure 6. Per-camera geometry coverage at frame 40 under the two target rig configurations. Dashed lines mark the seven-camera means, 94.07% for rig R1 and 96.66% for rig R2.

Table 8. Seven-camera geometry audit at key frame 40 under both target rigs. No target-view ground truth exists for either.

Target camera	R1 cov. (%)	R1 unk. (%)	R1 time (s)	R2 cov. (%)	R2 unk. (%)	R2 time (s)
Front Wide	98.83	1.19	15.02	99.39	0.61	8.69
Front Narrow	88.76	11.28	30.75	89.84	10.19	17.01
Front Left	97.67	2.34	13.03	98.96	1.05	8.49
Front Right	97.27	2.73	15.97	98.63	1.37	12.62
Back Left	95.07	4.95	20.31	98.38	1.63	11.02
Back Right	89.49	10.51	25.54	95.55	4.46	22.22
Back	91.43	8.60	20.19	95.89	4.11	10.91
Mean	94.07	5.94	20.12	96.66	3.34	12.99
Worst camera	88.76	-	-	89.84	-	-

Both fractions use the same denominator, the full frame of Table 6. They are not exact complements because they are read at different stages -- coverage after the geometry composition, unknown from the geometry export before any completion -- so a pixel resolved between the two reads appears in neither count; across the fourteen audits the largest resulting departure from 100% is 0.04 percentage points. Means are computed from the unrounded values, so they need not equal the mean of the displayed rows. Time is single-camera wall-clock render time.

The 2.6-point coverage difference between the rigs is why we report both rather than the more favourable one. R1 adds forward translation and pitch to R2's vertical displacement, and those extra degrees of freedom expose more disoccluded surface — unevenly, since the back-right camera loses 6.1 points and the back camera 4.5 while the front wide loses 0.6.

Coverage varies from 88.76% to 98.83% under R1 and from 89.84% to 99.39% under R2. A long focal length magnifies the image-space displacement produced by a given metric translation, which predicts that a narrow-field camera sweeps across more of its scene and exposes proportionally more disocclusion (Section 7.2). Figure 7 tests that prediction and qualifies it. The front narrow camera, which we also call the telephoto because its 29.3 degree field is what distinguishes it, is indeed the worst camera under both rigs, at 88.76% and 89.84%, and over all seven cameras coverage correlates with log focal length at $r = -0.70$ under R1 and -0.92 under R2. The focal length here is scaled to a common 1920 px raster, which matters: coverage is a fraction of each camera's own frame, fractional displacement goes as f/W , and the two front cameras are 3840 px wide while the surrounds are 1920. Correlating against unscaled f_x instead gives -0.51 and -0.84 , and makes the relationship appear to vanish when the telephoto is removed. Against the scaled quantity it does not vanish: the remaining six cameras give -0.53 and -0.62 . That is a weaker

relationship carried by more than one camera, not a single structural outlier. Six points on one frame still cannot establish a trend, and we report the association without claiming one.

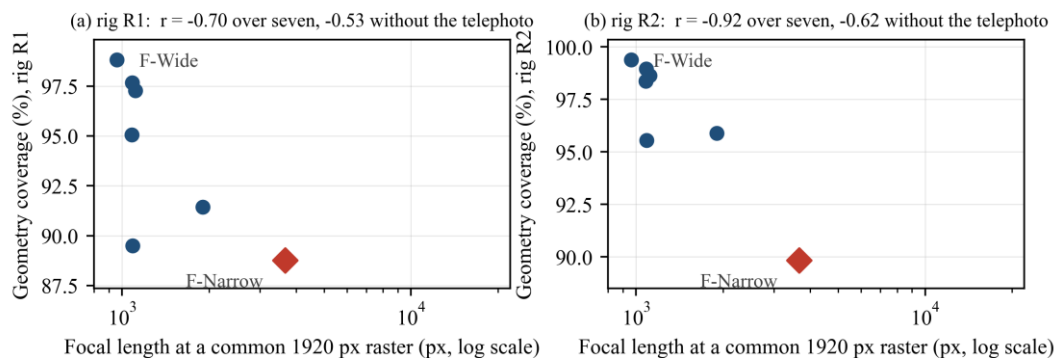


Figure 7. Geometry coverage against focal length scaled to a common 1920 px raster, per camera, under both target rigs. The telephoto is the red diamond. Over seven cameras the correlation with log focal length is -0.70 under R1 and -0.92 under R2, and -0.53 and -0.62 with the telephoto removed, so the relationship survives the outlier rather than depending on it.

Figure 13 shows which pixels each camera could not source at all, in the pipeline's own exported map. Figures 8 to 12 carry the same branch through its other exports: one camera stage by stage, the recorded reason each unsourced pixel stayed unsourced, the domain over which the rigid transform is defined, the exported depth, and depth beside the image it belongs to. The telephoto carries a single large unsourced region and the back camera two smaller ones, while the four surround cameras are almost solid. This is the same ranking Table 8 gives numerically, shown as the pixels it refers to.

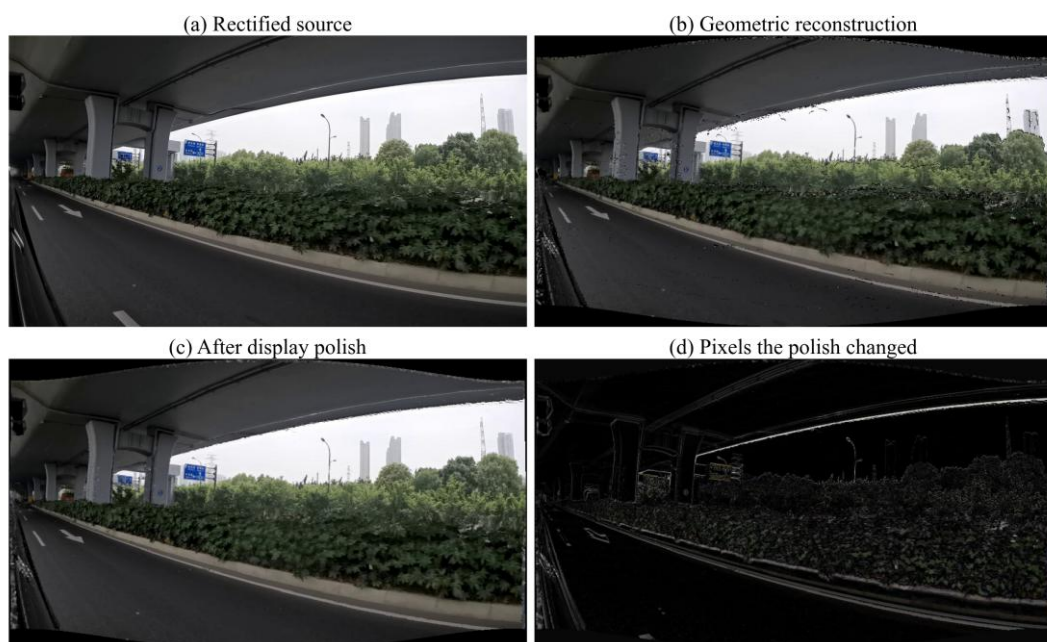


Figure 8. The back-left camera through the geometry branch in its own exported artefacts: (a) rectified source, (b) geometric reconstruction, (c) after the display-only polish, (d) the pixels the polish changed. Panel (d) is the visual form of the locked-pixel guarantee: the polish touches no pixel labelled observed.

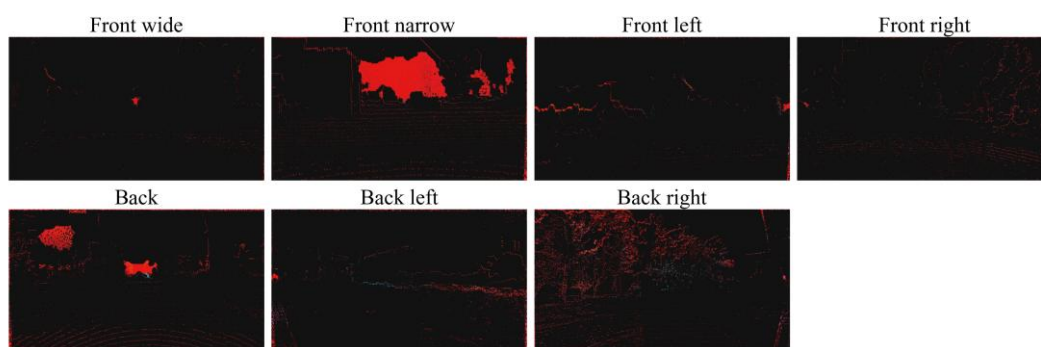


Figure 9. Unknown-reason map for each camera under rig R2. The pipeline records why each un-sourced pixel could not be filled rather than only that it could not, which is what allows an unknown region to be audited instead of merely counted.

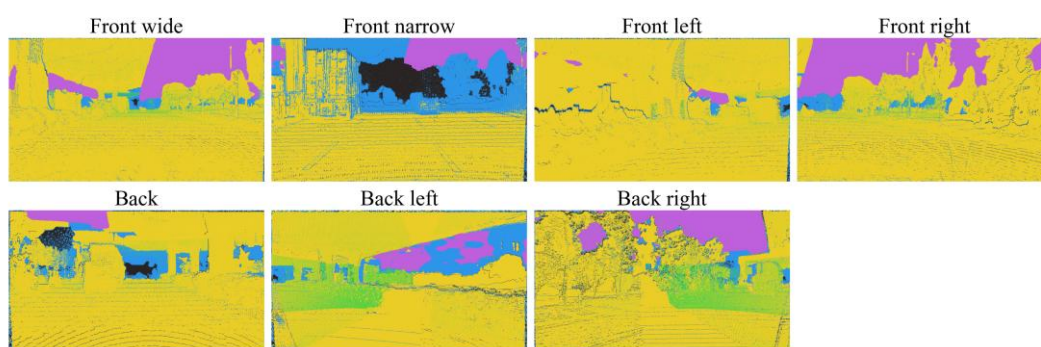


Figure 10. SE(3) domain map for each camera: the region over which the rigid rig transform is defined for that view. Regions outside it cannot be reached by the shared vehicle transform and are never claimed as observed.

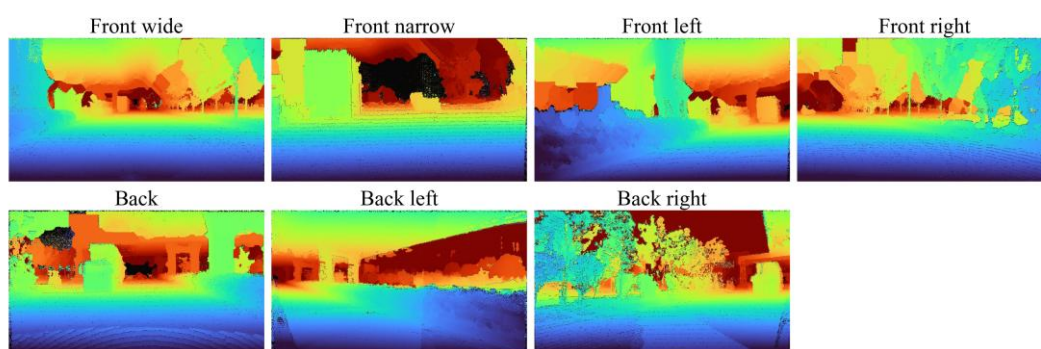


Figure 11. Exported reconstruction depth for each camera under rig R2. Depth is written only by the geometry path; no visual-only stage may modify it.

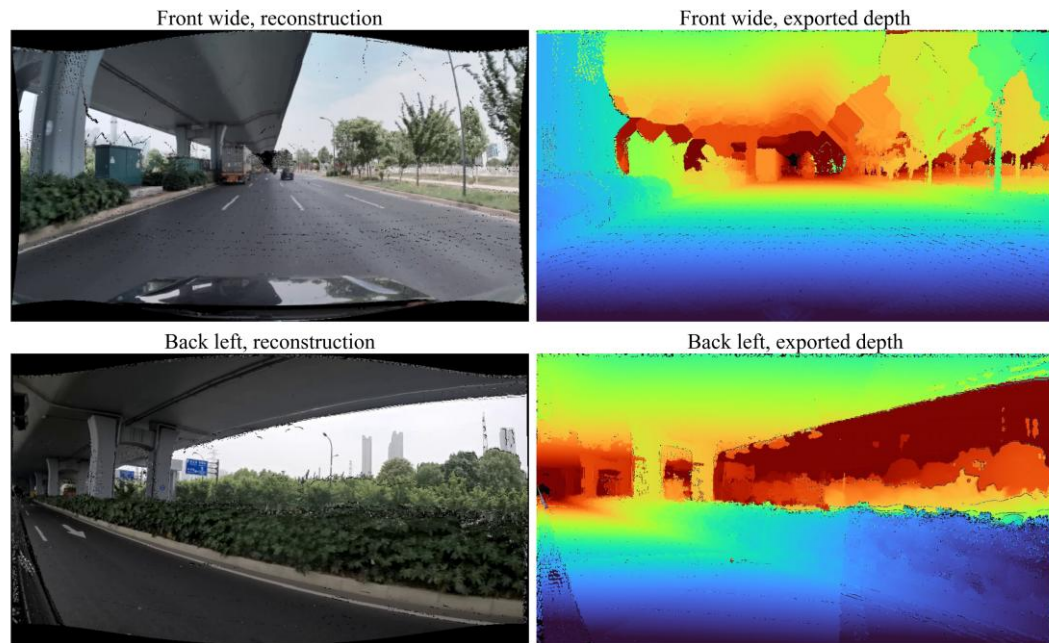


Figure 12. Reconstruction and its exported depth, side by side, for the front wide and back left cameras. The pairing shows the depth export is registered to the delivered image rather than to the source.

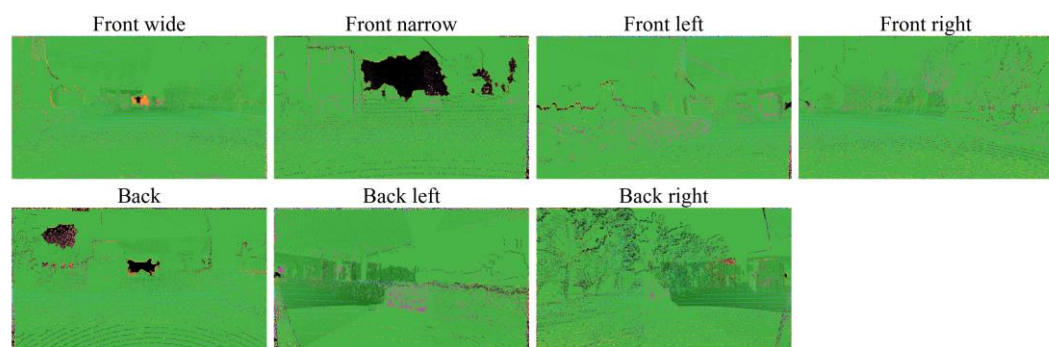


Figure 13. Hybrid source map for each camera under rig R2, as exported by the pipeline. Green marks a pixel written from a geometric source and black a pixel no source could supply. Nothing is redrawn; these are the delivered diagnostic rasters.

6.2. Hole-Free and Temporal Audits

A separate hole-free audit, Table 9 and Figure 14, concerns what is delivered for display rather than what was reconstructed. After the display-only completion all seven cameras reach 100% final visual coverage and pass the structural artefact test, with 388,533 pixels replaced by visual fill, 279 of them suspect near-black and no near-black component larger than 3 pixels against an admissible threshold of 9. That 100% claims only that the delivered image has no hole and no structural black artefact, not that the whole image was observed: the two differ by exactly those 388,533 pixels, which is why geometric acceptance is evaluated on the unpolished reconstruction.

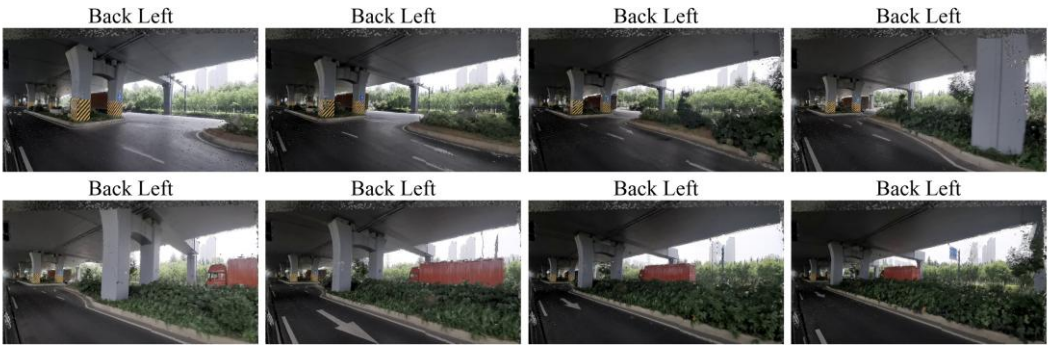


Figure 14. Delivered hole-free display images from the visual audit of this section. Every camera reaches 100% final visual coverage; these are the images that claim covers, and the geometric acceptance of Table 8 is evaluated on the unpolished reconstruction rather than on these.

Table 9. Hole-free visual audit for rig R1 at frame 40, concerning the delivered display image rather than the geometry export.

Target camera	Visual cov. (%)	Pixels replaced	Suspect near-black	Largest component (px)
Front Wide	100.00	21,056	8	1
Front Narrow	100.00	81,734	37	3
Front Left	100.00	24,669	53	2
Front Right	100.00	41,883	9	2
Back Left	100.00	44,116	117	2
Back Right	100.00	103,190	33	2
Back	100.00	71,885	22	2
Total / all cameras	100.00	388,533	279	3

Thresholds: near-black 8/255, change 8/255, maximum admissible component 9 px. All seven cameras pass. A 100% visual coverage means no hole in the delivered image, not that every pixel was observed.

Table 10 and Figure 15 report the geometry branch over 20 consecutive key frames for one camera under R1, each reconstructed independently: mean coverage 94.42%, standard deviation 0.44 percentage points, peak-to-peak spread 1.75 points between frames 31 and 39.

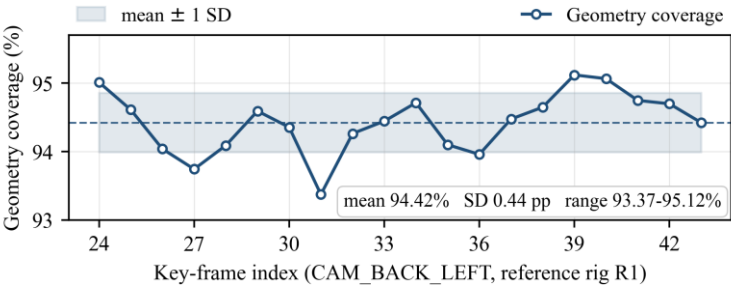


Figure 15. Geometry coverage over 20 consecutive key frames for the back-left camera under rig R1.

Table 10. Framewise geometry-coverage variation over 20 consecutive key frames (24-43) for CAM_BACK_LEFT under rig R1, each reconstructed independently with no temporal smoothing.

Statistic over frames 24-43	Geometry coverage (%)	Render time per frame (s)
Mean	94.42	17.53
Standard deviation	0.44	1.88

Statistic over frames 24-43	Geometry coverage (%)	Render time per frame (s)
Minimum (frame 31)	93.37	14.92
Maximum (frame 39)	95.12	21.88
Peak-to-peak spread	1.75	6.96

Each frame is reconstructed once, so the 0.44-point standard deviation cannot be separated into scene content and reconstruction instability; that would need repeated runs on a fixed input. It is not a substitute for a temporal-consistency metric, which remains unmeasured.

The narrowness of the band is a useful negative result: with no temporal filtering, per-frame coverage is driven by what the vehicle passes rather than run-to-run instability. It should not be over-read — coverage stability says nothing about whether a surface's colour flickers between frames, which is what a video consumer cares about, and that metric remains unmeasured.

A second temporal question can be put to the delivered video, and its answer is a caution rather than a result. The release includes 120 fps sequences of 4,741 frames that the manifest declares built from 80 real anchors by timestamp-aware linear interpolation, and Figure 16 tests that declaration against the pixels. Interpolation makes the frame-to-frame difference constant inside an anchor interval and discontinuous only at the anchors, which is what the pixels show: anchor crossings are 2.7 to 3.3 times the change between adjacent interpolated frames, and interval identity explains 32% to 76% of the variance, the residual consistent with H.264 compression noise.

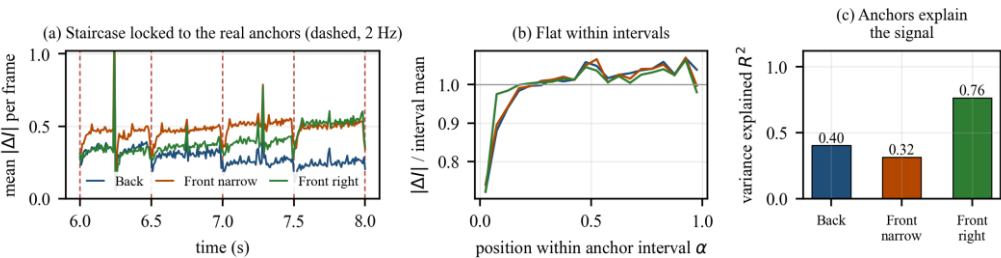


Figure 16. The delivered 120 fps video measured against its declared construction. (a) Frame-to-frame mean absolute difference over two seconds, real 2 Hz anchors marked; the signal is a staircase on the anchor grid. (b) Normalized by interval mean, it is flat inside each interval, as exact linear interpolation requires. (c) Variance explained by interval identity. A caution, not a result: the video carries 2 Hz of evidence at 120 fps.

The anchors sit 0.500 s apart, a capture rate of 2.000 Hz across the 39.5 s sequence: 80 of the 4,741 displayed frames are anchored to a source acquisition instant, each still a reconstruction rather than a measured target image, and the other 98.3% interpolation. The video is a presentation format, and no claim about flicker, temporal coherence or dynamic-object handling can be drawn from it. Measuring temporal consistency would need per-frame inference at 2 Hz under a flicker or warping-consistency criterion; no protocol in Table 7 covers it, which we note as a gap in that list.

6.3. Shifted-Pose Evaluation

Table 11 reports the aggregate learned-branch results over the 21 shifted-pose cases and Figure 17 the trend with per-camera spread. Observed coverage falls monotonically with the perturbation band, from 99.32% to 98.59% to 96.79%, overall 98.24%; gate filling falls from 54.86% to 41.49% to 38.34%, and the generated fraction rises from 0.37% to 0.82% to 2.10%, overall 44.90% and 1.10%. Every aggregate in Tables 11 to 13 is an unweighted mean over per-camera ratios, and the choice is consequential for the two metrics whose denominator is the hole rather than the frame. Observed coverage and generated fraction are frame-denominated, and all 21 cases share the same 1920 x 1080 raster, so for those the

unweighted and pooled values coincide. Gate fill and vehicle risk are hole-denominated and hole counts differ case by case, so for those they do not: pooled gate fill is 45.39%, 41.57% and 34.48% by band and 37.77% overall, against the 54.86%, 41.49%, 38.34% and 44.90% reported here. The two answer different questions -- the unweighted mean asks how the gate behaves on a typical camera, the pooled rate how it behaves over all hole pixels -- and every gate-fill and vehicle-risk figure in this article is the unweighted mean.

Table 11. A-ONLY shifted-pose proxy metrics at frame 73, averaged over seven cameras per band. The first three rows aggregate seven cases each; the last two summarize all 21.

Perturbation band	Observed cov. (%)	Gate fill of hole (%)	Generated / image (%)	Vehicle risk in hole (%)	Locked px changed
5-10 cm	99.32	54.86	0.37	5.18	0
10-20 cm	98.59	41.49	0.82	4.89	0
20-50 cm	96.79	38.34	2.10	4.76	0
All 21 cases	98.24	44.90	1.10	4.94	0
Worst value per column	92.19	11.70	5.67	28.00	0

Gate fill is a fraction of the hole; generated fraction is a fraction of the image. The last row takes each column independently, so its values need not come from one case.

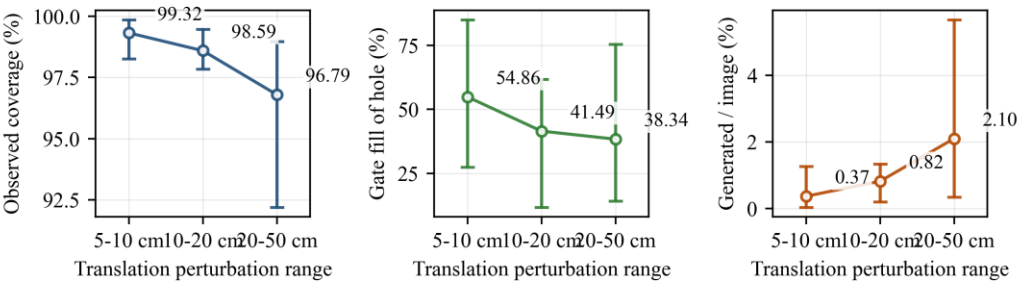


Figure 17. Aggregate A-ONLY behaviour against the translation perturbation band, at frame 73. Markers are the seven-camera mean, whiskers the per-camera range. Observed coverage falls and generated fraction rises with baseline, and the widening whiskers show the spread across cameras growing faster than the mean moves.

Figure 18 shows why a band average is the wrong summary for this quantity. Within a single band the seven cameras differ in gate fill by 57.63, 49.99 and 61.24 percentage points, while the three band means span only 16.52. The baseline moves the mean; which camera is being reconstructed moves the value several times further.

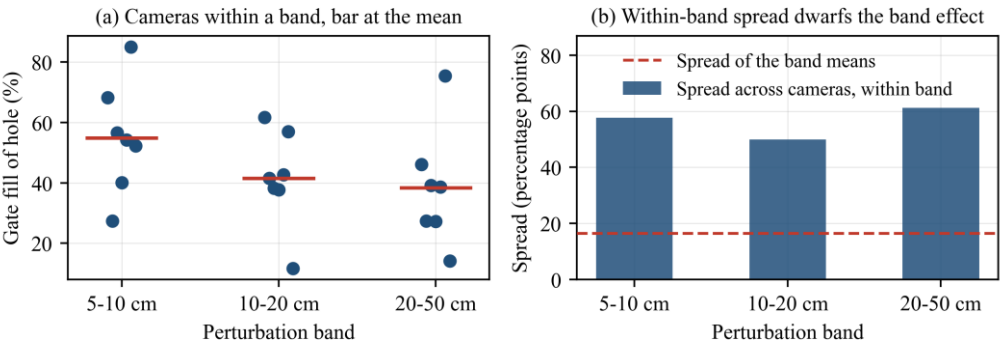


Figure 18. Gate acceptance within and across perturbation bands.

Figure 19 shows what the widest perturbation band actually delivers, for every camera. These are the exported images themselves rather than a summary of them, and they are included because a coverage figure of 96.79% does not tell a reader what the remaining 3.21% looks like.

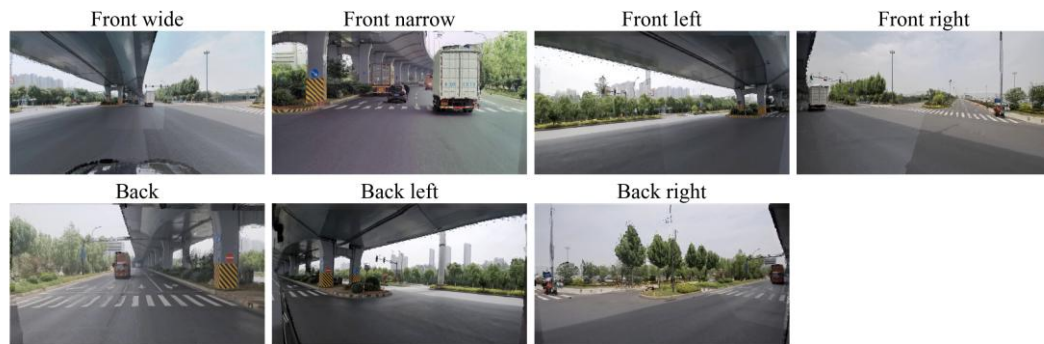


Figure 19. Delivered A-ONLY output at the 20-50 cm band for all seven cameras, at the exported mask resolution. The band is the hardest of the three and the one where the generated fraction is largest.

The three trends are coupled and behave as intended: a larger displacement exposes more unobserved surface, so the gate accepts less and more falls through to completion. A declining gate-fill rate is consistent with a discriminating gate rather than degradation, but without a held-out reference we can measure neither precision nor recall; this protocol verifies provenance and internal consistency, not gate correctness.

The widening whiskers matter too: at 5-10 cm the cameras span 98.26% to 99.86%, a range of 1.6 points, while at 20-50 cm they span 92.19% to 98.97%, a range of 6.8. Degradation is strongly camera-dependent and the mean becomes less representative as the baseline grows.

6.4. Per-Camera Analysis

Table 12 and Figures 20 to 22 decompose the same 21 cases by camera, and Figure 21 sets the learned branch against focal length. Two cameras dominate the risk profile, and neither is visible in the aggregate.

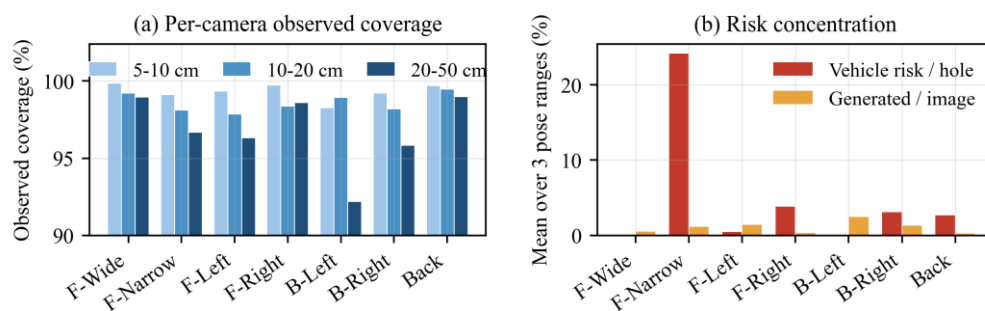


Figure 20. Per-camera decomposition of the 21 shifted-pose cases. (a) Observed coverage for each camera in each perturbation band; the back-left camera degrades fastest. (b) Mean vehicle-risk fraction of the hole set and mean generated fraction. The front narrow camera carries a vehicle-risk fraction 4.9 times the seven-camera mean, which no aggregate report would show.

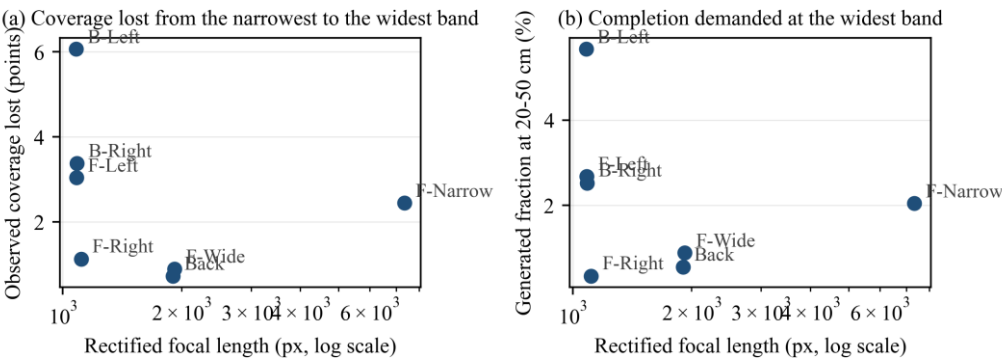


Figure 21. Learned-branch behaviour against focal length, scaled to a common 1920 px raster. (a) Observed coverage lost between the narrowest and widest perturbation band. (b) Generated fraction at the widest band. Neither panel shows the clean ordering by focal length that the geometry branch shows in Figure 7: the back-left camera loses most on both counts while sitting at the short-focal end.

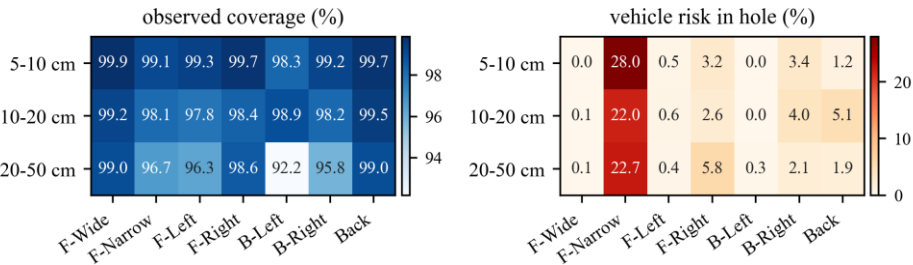


Figure 22. Per-camera, per-band observed coverage (left) and vehicle-risk fraction of the hole set (right). Coverage degrades smoothly and broadly with baseline, whereas vehicle risk is concentrated almost entirely in one column.

Figure 23 traces each camera through the three bands rather than reporting its endpoints. The paths are not parallel: the back-left camera loses 6.06 points of observed coverage between the narrowest and widest band while the back camera loses 0.73, and the two cross in gate acceptance. A per-camera mean over bands, or a per-band mean over cameras, hides this in either direction.

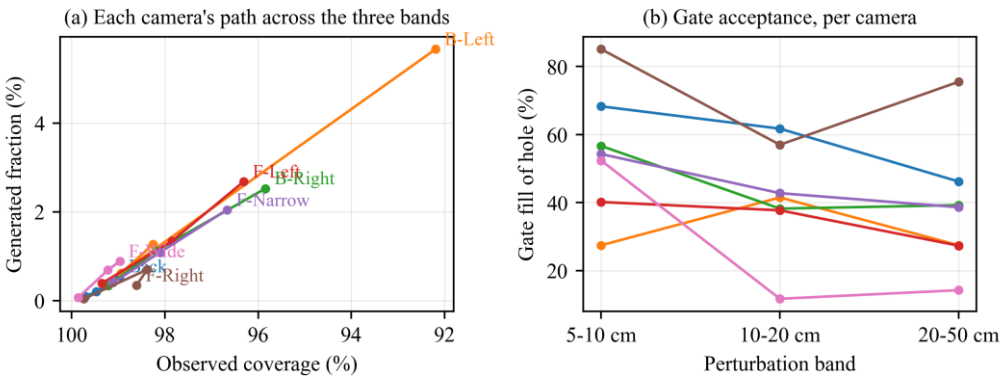


Figure 23. Each camera traced across the three perturbation bands.

Figure 24 shows where that risk actually falls. The mask is nearly empty for six cameras and carries substantial structure for the telephoto, which is the 24.23% against a 4.94% mean of Table 12 rendered as the pixels it counts.

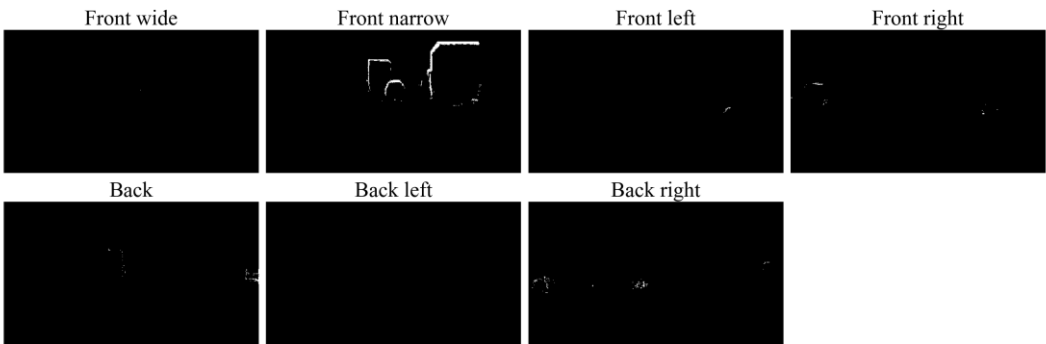


Figure 24. Vehicle-extension risk masks at the 10-20 cm band, all seven cameras, as exported. White marks a hole pixel falling on a region identified as a vehicle. The telephoto panel carries most of the rig's risk, and the other six are nearly empty.

Table 12. Per-camera A-ONLY profile, averaged over the three perturbation bands. The seven-camera mean of Table 11 hides the concentration of risk visible here.

Target camera	Observed cov. (%)	Gate fill of hole (%)	Generated / image (%)	Vehicle risk in hole (%)	Worst-band obs. cov. (%)
Front Wide	99.35	26.09	0.55	0.04	98.97
Front Narrow	97.96	45.19	1.18	24.23	96.67
Front Left	97.84	35.04	1.47	0.50	96.31
Front Right	98.91	72.49	0.36	3.86	98.38
Back Left	96.46	32.10	2.52	0.11	92.19
Back Right	97.75	44.68	1.33	3.15	95.84
Back	99.38	58.70	0.28	2.73	98.97

Front Narrow carries roughly five times the seven-camera mean vehicle risk; Back Left has the lowest coverage and the largest generated fraction.

Three observations follow from the per-case rows behind these aggregates. The ranking of cameras by observed coverage is not stable across bands — the back-left camera is last at 5-10 cm, third at 10-20 cm and last again at 20-50 cm — and only the front wide camera holds a top-two position throughout. Gate fill varies far more across cameras within a band, by 50.0 to 61.2 percentage points, than the band means vary across bands, by 16.5. And the locked-pixel column is zero in every row.

The front narrow camera has a mean vehicle-risk fraction inside holes of 24.23%, against a seven-camera mean of 4.94% and a worst case of 28.00% — 4.9 times the seven-camera mean, and nearly a quarter of its holes falling on regions identified as vehicles, so the completion module is repeatedly asked to invent a traffic participant rather than road surface. Section 7.2 argues this needs instance-aware handling, not a stronger generator.

Figures 25 and 26 follow the telephoto across every band it was evaluated on, in delivered output and in risk mask. The back-left camera shows the other failure mode: the lowest mean observed coverage at 96.46%, the largest mean generated fraction at 2.52% and the worst case at 92.19% coverage with 5.67% generated, yet a vehicle risk of 0.11%, the second-lowest of the seven. The two fail in orthogonal ways, and a single scalar would rank them alike and mislead deployment planning.



Figure 25. The telephoto camera's delivered output across all three perturbation bands, the camera that carries the highest vehicle-risk fraction, shown at every band it was evaluated on.

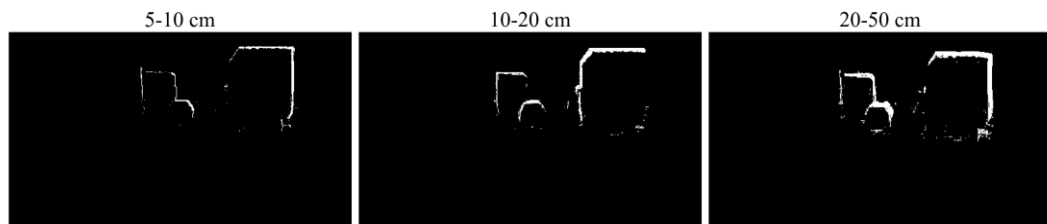


Figure 26. Vehicle-extension risk masks for the telephoto across the three bands. The risk is present at every band rather than appearing only at the widest, so it tracks the camera rather than the size of the perturbation.

This concentration coincides with a measurable geometric asymmetry, though one sequence cannot separate the optics from the scene content each camera happens to face. Figure 27 shows that cloud reprojected into each camera separately. Reprojecting the 313,196-point target cloud through each calibrated camera gives the angular LiDAR support each receives, and the telephoto is an extreme outlier: its 29.3 degree field at 8.3 megapixels collects 2,053 returns per megapixel of its native raster, 5.0 times fewer than the next sparsest. The ratio is a property of the raster it is computed on: at the common 1920 x 1080 evaluation raster the same returns give 8,214 per megapixel against 18,435 for the next sparsest, a factor of 2.2. The ranking survives the change of raster and the magnitude does not, and per solid angle, which is how a rotating scanner actually distributes returns, the telephoto is not starved at all -- it is the densest camera on the rig. What the per-pixel figure measures is how many pixels must be interpolated per return, which is the quantity that matters for filling a raster (Figure 28; Figure 29 shows the same fact visually).

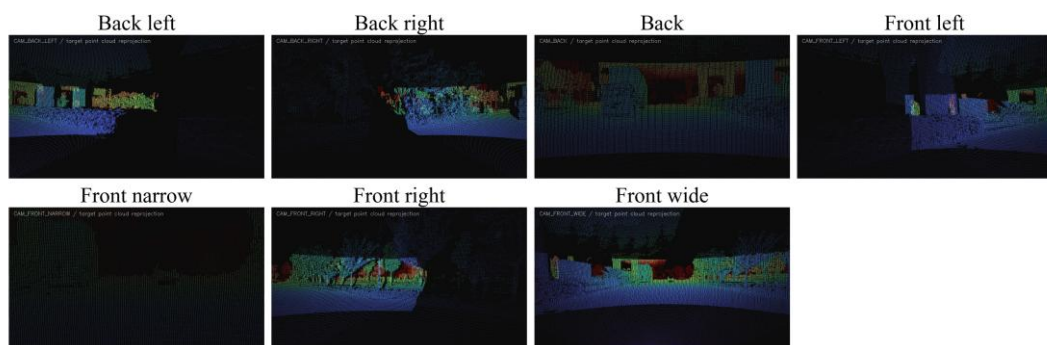


Figure 27. The target-frame LiDAR cloud reprojected into each calibrated camera individually. The telephoto panel is visibly the sparsest per pixel, which is the per-camera support that Figure 28 reports numerically.

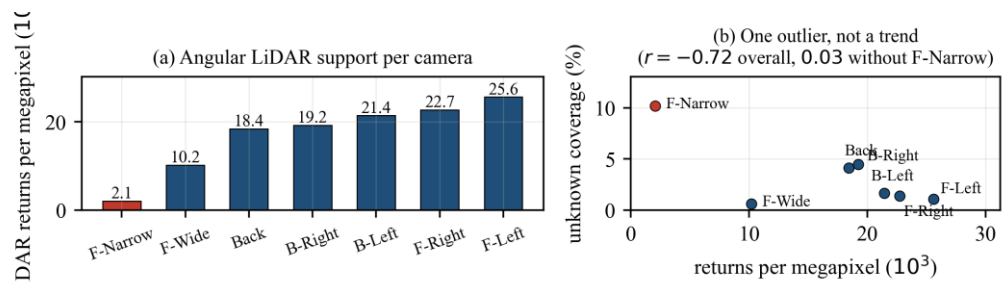


Figure 28. Angular LiDAR support per camera, from reprojecting the 313,196-point target cloud through each calibrated camera. (a) Returns per megapixel of each camera's native raster; the telephoto receives 5.0 times fewer than the next sparsest, and 2.2 times fewer at the common evaluation raster. (b) Support against unknown coverage, counted on the 1920 x 1080 audit raster: $r = -0.81$ over all seven and -0.69 without the telephoto, so the relationship does not rest on that camera alone.

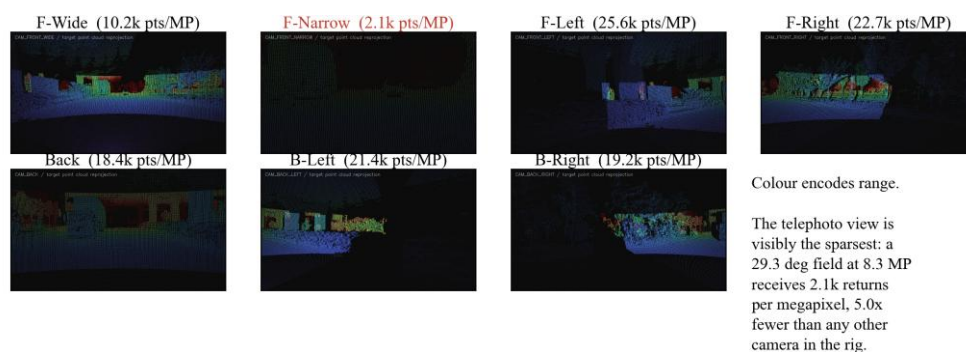


Figure 29. The target-frame LiDAR cloud reprojected into all seven calibrated cameras, colour encoding range. The telephoto panel is visibly the sparsest, which is the same fact Figure 28 states numerically; the reprojection also serves as a calibration sanity check, since a wrong extrinsic or array ordering would place returns off the structures they belong to.

How far this generalizes needs care. Counting returns per megapixel on the raster the audit actually used, 1920 x 1080 for every camera under R2, the correlation between support and unknown coverage is -0.81 over all seven and -0.69 with the telephoto removed. Counting them instead on each camera's native raster gives -0.72 and $+0.03$, which would suggest the relationship rests on one camera; it does not, and the difference is an artefact of mixing rasters rather than a finding. Six points on one frame remain too few to establish a trend, so we report the association and not a law. One design statement survives independently of any correlation: a long-focal-length camera samples a narrow solid angle at high pixel density, so a rotating LiDAR of fixed angular resolution necessarily gives it the least evidence per pixel in the rig, which predicts where completion will do the most work.

The delivered masks can be checked against the results table. Re-measuring the 21 exported vehicle-extension rasters from the PNG files against the product of the published risk and hole fractions gives a correlation of 0.9969 (Figure 30). They agree in proportion, not absolute value — the fitted slope is 2.38, because the masks are exported at 512 x 288 with dilation while the metric uses the native raster. Both come from the same pipeline, so this confirms they describe the same pixels and is not independent validation. Panel (b) shows the telephoto spanning 22.0% to 28.0% of its hole set across the bands, while no other camera exceeds 5.8%.

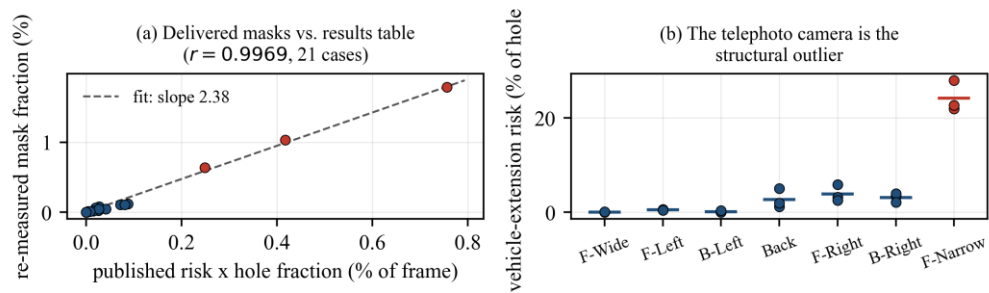


Figure 30. Consistency check between the delivered risk masks and the results table. (a) The 21 exported vehicle-extension rasters, re-measured from the PNG files, against the product of the published risk and hole fractions; they agree in proportion ($r = 0.9969$) with a scale factor set by the export resolution.

Both guarantees hold for every case at once. The partition closes to within $1.5e-14$ points, the rounding of double-precision arithmetic and nine orders of magnitude inside the $5e-6$ tolerance, and the count of locked observed pixels whose value changed is zero in all 21 cases. Neither is a statistical result: both follow from the compositing code, and they are a regression test on that code rather than a measurement of the scene.

6.5. The Evidence Budget and Locked-Pixel Compliance

Table 13 and Figure 31 express the same results as a budget over the delivered frame. At every perturbation band the observed, gate-recovered and generated shares sum to 100.000%.

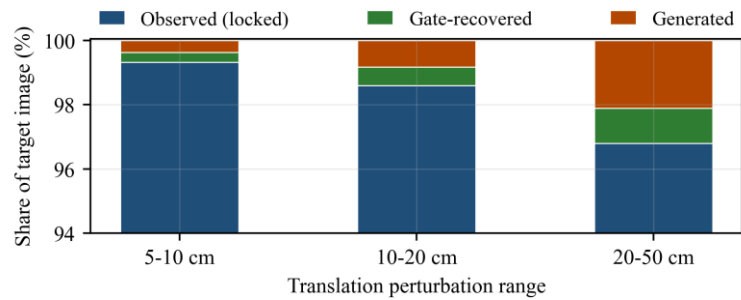


Figure 31. Evidence budget of the delivered A-ONLY frame at each perturbation band. The three provenance classes partition the image exactly. The observed share contracts from 99.32% to 96.79% as the baseline grows, and the growth is absorbed by the gate and generated classes in a ratio that shifts steadily towards generation.

Table 13. Evidence budget of the delivered A-ONLY image as a share of the target frame. The three provenance classes are mutually exclusive and exhaustive.

Perturbation band	Observed and locked (%)	Gate-recovered (%)	Generated (%)	Sum (%)
5-10 cm	99.320	0.309	0.371	100.000
10-20 cm	98.592	0.585	0.823	100.000
20-50 cm	96.795	1.105	2.100	100.000

The columns sum to 100% at every band: each delivered pixel is observed, gate-recovered or generated. The unknown mask is empty in the delivered RGB but still exported.

This closure is the arithmetic statement of the contract: no fourth category, no unaccounted remainder. Between the narrowest and widest bands the observed share contracts by 2.53 points, 0.80 absorbed by the gate and 1.73 falling through to generation, so gate recovery relative to generation falls from 0.83 to 0.53.

In all 21 cases the number of locked observed pixels whose value changed is exactly zero, and Figure 32 maps where the polish did act. The guarantee is structural rather than statistical — the observed region is removed from the modification mask of Equation (20) by construction, so a non-zero count would signal an implementation defect — and we report it as a regression test to be re-checked on every future configuration. The count is taken on the rectified internal raster, which is where Section 3.4 defines the lock. Export redistorts that raster, and resampling interpolates: a delivered pixel near a class boundary can mix observed and recovered contributions, so byte equality is not claimed for the redistorted image and is not measured there. The same resampling raises the question of what label a delivered pixel carries. The contract answers with a conservative rule: a delivered pixel receiving any contribution from a generated pixel is itself generated, and one mixing observed and gate contributions is gate, so a label can only move toward lower evidence. Categorical labels are remapped by nearest neighbour while RGB is interpolated, which is why the rule has to be conservative: the two resamplings need not agree at a class boundary. What we do not report is how many pixels that disagreement touches. Figure 33 shows the two rasters involved. Measuring it is protocol P8 of Table 7, and it is not a byte comparison between the two rasters -- they are different domains and would differ almost everywhere -- but a comparison of the delivered image against the locked reference carried through the same remap, counting cross-class contamination at boundaries.

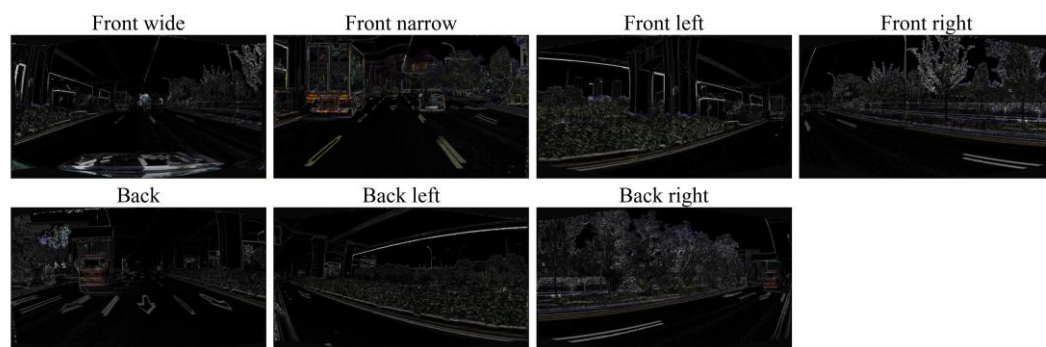


Figure 32. Pixels changed by the display polish, per camera. The polish is confined to the gate-recovered and generated classes by construction, and these maps are the exported evidence of that confinement.



Figure 33. The two exported rasters for one case: the delivered image and the mask raster it is audited at. The masks are exported at the lower resolution, which is the scale factor of 2.38 that Figure 30 reports between re-measured and published risk, and the resampling this section's labelling rule has to survive.

One further check separates two questions the compliance numbers above run together: whether the pipeline behaved, and whether the rules could be broken without anyone noticing. A reference implementation of the specification alone -- Equation (7),

Equation (20), the Section 4.11 transitions and the export rule, written from this article and sharing no code with the pipeline -- satisfies six invariants over 20,000 randomized frames, and breaking any one rule makes the matching invariant fail on more than 91% of them, so the invariants are falsifiable rather than vacuous. This says the contract is internally coherent and that violations are detectable in principle. It says nothing about whether GCR-NVS implements it, which is a different claim needing the pipeline and is protocol P5 of Table 7. The checker is in the released repository and needs neither the sequence nor a checkpoint.

6.6. Qualitative Inspection

Figure 34 traces one case through the whole pipeline using real intermediate artefacts: rectified source RGB, dense depth candidate, semantic feature projection, structure-constraint audit, target surface depth, visibility and provenance, gate output, final reconstruction, absolute difference against the rectified source, and the redistorted delivery image. Nothing in that delivered image shows which pixels are measurement and which are completion, which is the argument for exporting the masks alongside it. Its absolute-difference panel compares the rectified source with the rectified final reconstruction; because the target camera has moved it shows parallax as well as error, so it locates tearing and ghosting but yields no fidelity score.

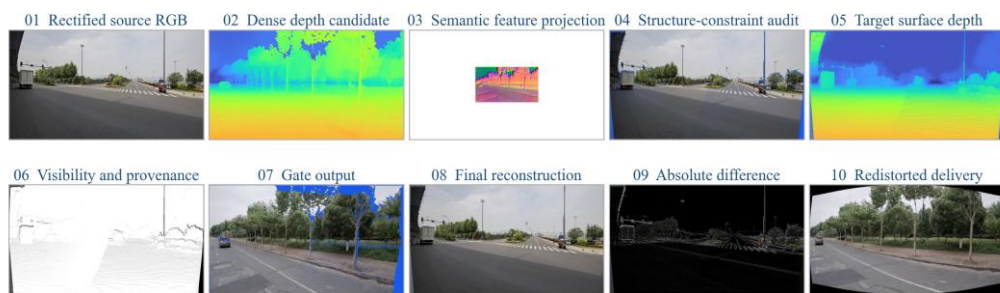


Figure 34. A-ONLY process trace for sequence 2026-05-22-10-25-21, frame 73, 10-20 cm band, front-right camera. Panels are cropped from the inference artefacts rather than redrawn, read left to right and top to bottom in the order Section 6.6 lists them. Panel 03 fills little of its frame because the semantic feature projection is computed at reduced resolution and is shown at its own scale.

6.7. On the Absence of an Ablation

Section 5.5 records why no component-wise ablation is reported. What the results support is narrower: the composed system holds the contract exactly and its coverage degrades predictably with baseline. Protocols P5 and P6 of Table 7 would close this without new collection.

7. Discussion

7.1. Large-Baseline Disocclusion Is an Observability Boundary, Not a Model Deficiency

When a target view exposes a surface absent from every source image and temporal neighbour, no geometric method can recover its true texture. This is a statement about the information content of the capture, not the capacity of the model, and the correct output is an unknown mask; any pixel written into it is a hypothesis however it was produced.

The generated fraction grows from 0.37% to 2.10% across the bands. It tracks this boundary without measuring it, since it also moves with depth error, gate thresholds, source selection and completion policy: even at 20-50 cm an average of 96.79% of the frame stays supported by a calibrated observation and 2.10% is hypothesis. That ratio is favourable precisely because the displacement is small; extrapolating it to a genuine platform

change would be unjustified, since the relationship is scene-dependent and must be measured per configuration.

7.2. Focal Length, Camera Geometry and the Concentration of Risk

The front narrow camera is the worst performer on geometry coverage under both rigs and carries the highest vehicle-risk fraction, and two measured optical asymmetries accompany that deficit. Induced image-space displacement scales with focal length, and this camera has 3.8 times the focal length of the front wide, which shares its raster, and 3.3 to 3.4 times the surrounds once each focal length is scaled to a common image width (Table 1); its field of view is 2.8 times narrower than theirs, so the same rig motion sweeps it across more of its field of view while its narrow field of view reduces the overlap neighbours could supply.

The second asymmetry is on the input side, and Section 6.4 gives its numbers. A rotating LiDAR of fixed angular resolution places returns by the solid angle a field subtends rather than by pixel count, so a narrow field at high pixel count collects fewer returns per pixel: the telephoto receives 2,053 per megapixel of its native raster against 10,175 to 25,596 for the rest of the rig, and 8,214 against 18,435 for the next sparsest once every camera is counted on the common evaluation raster. The direction is raster-independent and the magnitude is not, and per solid angle the telephoto is the densest camera on the rig rather than the sparsest, so this is a statement about pixels per return and not about the scanner neglecting that direction. So it needs the most interpolation per return and receives the least support per pixel, which is a rig-design consideration rather than a model one and is invisible to any aggregate metric.

Both asymmetries are measured; neither is shown to cause the deficit. One sequence cannot separate optics from the scene content this camera happens to face, and settling it needs a controlled sweep over focal length, output resolution and pose on a fixed scene. We report the association and the mechanism that would explain it, and stop there.

The consequence is direct: a seven-camera mean of 4.94% is comfortable, the 24.23% carried by one camera is not, and a deployment decision taken on the mean is taken on the wrong quantity. Per-camera reporting is mandatory rather than supplementary.

The remedy is not a stronger completion model. When a quarter of a camera's holes fall on vehicle regions the correct response is instance-aware handling: detect and track the instances, then recover them from a temporally adjacent observation or leave them explicitly unknown. Generating a plausible vehicle is worse than leaving the hole, because a fabricated traffic participant propagates silently into whatever consumes the corpus.

The back-right camera shows a related but distinct pattern, with the largest inter-rig coverage difference at 6.06 percentage points; its behaviour is dominated by rig geometry rather than focal length, since it sits where R1's forward and pitch components combine least favourably with the existing mounting angle.

7.3. Dynamics and Temporal Interpolation

The pipeline assumes a static scene between the observations it fuses; moving vehicles, pedestrians, vegetation and the finite duration of a LiDAR sweep all violate that. Consistency gates, source priority and dynamic-region masks mitigate it, but instance-level scene flow is not estimated, so long-horizon fusion can ghost on movers. The demonstration video interpolates between real anchors 0.500 s apart, 80 of 4,741 displayed frames being evidence and 98.3% interpolation (Section 6.2); a reader judging temporal quality from it would be reading the interpolator.

7.4. Why Conventional Image Metrics Are Insufficient Here

In the no-target-ground-truth setting PSNR, SSIM [49] and LPIPS [50] are either undefined or actively misleading, and it is worth being precise about which.

They are undefined without a target-view reference, the situation for every result in Section 6, and misleading when one is improvised from the same-named source camera: by Equation (19) that camera dominates the fusion weight at small baseline, so a system that copies it scores well and one that correctly renders the induced parallax scores worse. The metric would reward the behaviour the system exists to avoid.

What we report instead is an evidence profile: coverage with a traceable source, gate acceptance under an explicit cycle test, bounded generated fraction, exact locked-pixel compliance and per-camera risk. It is weaker than a photometric score in not measuring how correct the delivered colour is, and stronger in that every quantity has a defined support and a stated measurement status. Once a double-rig collection exists the conventional metrics become defined and should be added alongside the profile, not substituted for it.

7.5. The Same-Name Camera Caveat

The 21 shifted-pose cases are Task B: the same-name source camera is in the input, a translation-only perturbation leaves its optical axis unchanged, and Equation (19) gives it a dominant weight. They demonstrate correct pose handling, a consistent extrinsic chain, continuous motion, a discriminating gate and an enforced contract. They do not demonstrate synthesis of a viewpoint the rig cannot see, so the 98.24% observed coverage must never be quoted as evidence of that capability. P1 of Table 7 needs no new hardware and admits an exact photometric ground truth.

7.6. Implications for L4 Logistics Vehicle Programmes

Three suggestions follow for a fleet operator, offered as hypotheses rather than conclusions: they rest on one sequence, two parameterized rigs and a single learned-branch key frame, and the threats of Section 7.7 apply to all three. **First**, the target rig should be a versioned configuration artefact, not a constant in code; the 2.6-point difference between two rigs separated by a modest displacement and a two-degree pitch shows how quickly results become incomparable, so a metric quoted without a rig identifier is unusable. This follows from the paired audit of Table 8 and is the best supported of the three. **Second**, acceptance criteria should be written in terms of evidence rather than appearance: a hole-free requirement is trivially met by a generative model, whereas exported provenance masks, zero locked-pixel modification and stated bounds on coverage and generated fraction are checkable and not satisfiable by hallucination. **Third**, per-camera thresholds likely need setting from each camera's geometry, since a telephoto and a wide surround did not degrade alike under the same motion.

Three parts of this work are portable to a team with its own multi-camera corpus, independently of our pipeline: the provenance contract of Section 3.4 and Table 3, which specifies what a reprojection tool must export and what each stage may write; the evidence profile of Section 5.4, which needs only masks such a tool already computes; and the per-camera risk diagnostic of Section 6.4.

7.7. Threats to Validity

- **Single sequence, single condition.** All reported numbers come from one sequence, in one location, under one set of lighting and weather conditions, on one road type. Coverage is a function of scene geometry and appearance: an urban canyon, a tunnel, an open highway, night, rain or low sun would each produce different values, and low-texture or specular conditions would additionally stress the plane sweep and the cycle gate in ways this sequence cannot reveal. The reported figures characterize the pipeline on this data, not the pipeline in general, and no variance estimate across conditions is available.
- **Single frame for the learned branch.** The 21 shifted-pose cases are 21 camera-pose combinations at one key frame, not 21 independent scenes. They share scene content,

and the per-camera variation reported in Section 6.4 partly reflects what each camera happens to be looking at in that frame.

- **Estimated rig parameters.** The LiDAR ground height, body-frame camera positions and vehicle dimensions used to construct the target rigs are documented team estimates, not measured values. They affect the absolute interpretation of the target geometry, though not the internal consistency of the comparison between the two rigs.
- **One point of sensitivity, not a curve.** The estimate above raises the obvious question of how much the reported coverage would move if a rig parameter were wrong, and the paired audit answers it at exactly one point. The two rigs differ in more than one respect, and the difference must be read accordingly. R1 sits 0.10 m forward, 0.35 m up and 2 degrees pitched from the source while R2 is approximately 0.20 m up, so their separation combines a forward translation, about 0.15 m of height, a rotation, and a change of delivery raster from 960 x 540 to 1920 x 1080. Mean geometry coverage differs by 2.6 points between them. That is a measured response to a known *configuration* change, not to a controlled vertical displacement, and no part of it can be attributed to height alone. The response is also strongly camera-dependent -- the back-right camera moves 6.06 points and the front wide 0.6 -- so no single ratio extrapolates to other magnitudes or to rotation. Bounding the estimates properly would need a sweep over rig parameters, which is a configuration change and compute rather than new collection, and which we have not run.
- **Proxy metrics throughout.** Every quantity in Section 6 is a coverage, provenance or consistency proxy. None is a fidelity measurement against an independent reference, and no combination of them establishes that the delivered colour is correct.
- **No independent geometric validation.** The depth field is constrained by LiDAR and evaluated on data that includes the constraining points. A held-out LiDAR subset is required before any statement about geometric accuracy can be made, and none is made here.

8. Reproducibility, Data Lineage and Ethical Data Use

8.1. Artefact Organization and Data Lineage

The implementation has two branches with separate manifests: a geometry branch that runs without any learned component, and an inference-only learned branch consuming the same calibration and rig configuration. A run records the source sequence and frame, target rig, resolution, seed, code revision, checkpoint hash, device, elapsed time and output paths. Optional visual models can be disabled without altering the geometric exports.

The inference configuration is fully specified, and we give the identifiers rather than pointing at a run report. The frozen project checkpoints are `t0_structure_rectified_v2_dinob14_320x180_c48_best.pt` for the structure-constraint network, `adapter_step_001000.pt` for the A-ONLY residual adapter and `anyup_multi_backbone.pth` for the feature upsampler; the frozen third-party models are DA3Metric-Large for dense metric depth, `dinov2_vitl14` for semantic features at a 364 x 644 input, and `dinov2_vitb14` as the structure network's own feature backbone. Section 4.5 gives the resulting parameter counts, obtained by instantiating the released modules rather than quoted from a specification.

Training is a different matter. The released package is inference-only: model definitions, frozen weights and the configuration that consumes them, but not the training scripts, losses, optimizer settings, schedules or data splits that produced those weights. No training happens during evaluation, so the evaluation path is deterministic given its inputs; but the release carries neither the road-collection data nor the checkpoints, and we cannot show that the evaluation frames were excluded from the training or checkpoint

selection of the frozen weights, because the splits that produced them are not ours to publish. Reproduction from this release alone is therefore not possible, and we do not claim it. We state the split explicitly, because a reproducibility claim that does not say which half it covers is worth little. The geometry branch is reproducible in principle from the released code given the sequence and the calibration: it carries no learned component, and Tables 8, 9 and 10 come from it. **The learned branch is a non-reproducible demonstration.** Its checkpoints, training data, losses, optimizer settings and selection splits are not ours to release, so Tables 11 to 13 and every A-ONLY figure should be read as a record of what one frozen configuration did, not as a result another group could regenerate and check. Retraining rather than rerunning would need material this artefact does not carry, and we prefer to say so than to imply otherwise.

Table 14 records the eleven lineage classes and, for each, the provenance label of Equation (7) it produces, whether it may export depth, and whether it is eligible for SE(3) optimization. The table is machine-readable in the artefact package and is the authority consulted at composition time rather than a post-hoc description of it. Its priority column also defines compositing order: a lower-priority class may write only where every higher-priority class has declined.

Table 14. Data lineage classes with the provenance label of Equation (7) each produces. Classes are processing stages, a finer granularity than the partition of Table 3; the priority column also defines compositing order.

Pri.	Class	Input and processing	Maps to Q(p)	Exports depth	SE(3) eligible	Trust note
1	LiDAR anchor	Calibrated projection and z-buffer	observed	Yes	Yes	Measured geometry
2	Current RGB-D	Rectified RGB with LiDAR-aligned dense depth	observed	Yes	Yes	Estimated depth; anchors restored
3	Temporal	Same-camera neighbouring frames, SE(3) warp and gates	gate	Yes	Conditional	Only after consistency gates
4	Multi-view	Adjacent cameras, rigid-rig reprojection, visibility gates	gate	Yes	Conditional	Fills geometry unknown only
5	TSDF	LiDAR-anchored charts, sparse zero-level rendering	gate	Yes	Conditional	Hard anchors remain exact
6	Small-hole patch	Guarded local fill bounded by known RGB	generated	No	No	Visual output only
7	Splatting appearance	Calibrated sequence RGB with trusted depth	generated	No	No	Visual output only
8	Directional background	Semantic sky consensus, rotation-only warp	generated	No	No	Visual output only
9	Masked inpainting	Unknown mask excluding directional fill and anchors	generated	No	No	Visual only; strict mask composite
10	Display polish	Impulse removal, bilateral blend, unsharp mask	none (transform)	No	No	Masked out of every observed pixel; anchors restored byte-exactly
11	Redistortion export	Pinhole-domain RGB and masks remapped with raw K, D	none (transform)	No	No	Display only; geometry stays in the rectified domain

Classes 1 to 5 may contribute to geometric exports; 6 to 11 may not, and none can alter a pixel labelled observed. Classes 10 and 11 carry no label because they transform an already-labelled image.

Two rows carry the safety weight. The LiDAR anchor row is the only class marked measured geometry and the only one restored byte-exactly after the polish as well as masked out of it. Every visual-only class exports no depth and is ineligible for SE(3), the machine-checkable form of the rule in Section 3.4.

Two reproduction hazards are worth flagging. The calibration array ordering of Section 4.2 yields plausible-looking but physically wrong reprojections when consumed in name order, and any geometric step performed in the distorted domain produces peripheral errors resembling mild miscalibration that a central-region check will not catch. Figure 29 guards against both: it reprojects the target-frame LiDAR cloud into all seven cameras, so either fault scatters returns off the structures they belong to, which is obvious in a reprojection and easy to miss in a coverage statistic.

The geometry-branch results in Tables 8, 9 and 10 reproduce without any learned checkpoint, because every learned module in Table 4 can be disabled without altering the geometric exports; they need only the source sequence, the calibration, the distortion coefficients and one target rig configuration. The learned-branch results additionally need the frozen structure and A-ONLY checkpoints, whose hashes are recorded in each run report.

8.2. Ethical and Compliance Position

The reference data are used under a challenge data-use agreement for development, self-evaluation, demonstration and review, and are not redistributed. Raw RGB, point clouds, calibration and sensitive derived data are excluded from any public repository, and licence plates, faces and sensitive geographic information are anonymized to the provider's requirements before release.

Two commitments are verifiable from the exported artefacts rather than resting on assertion: results are produced by batch execution of the released code, not manual per-frame substitution, and generated pixels are separated from measured geometry by the provenance and hole masks. The target vehicle parameters are a team-defined reference preset, not a manufacturer measurement; if a measured calibration becomes available the configuration must be versioned upward, results regenerated, and old and new not pooled.

9. Limitations and Future Work

9.1. Limitations

Section 7.7 states the four threats that bound every number in this paper: a single sequence and condition, a single key frame for the learned branch, estimated rather than measured rig parameters, and proxy metrics with no independent geometric validation. They are not repeated here. One further limitation is a property of the model rather than the evidence.

- **Modelling gaps.** Dynamic-object scene flow is not modelled at instance level, higher-resolution coarse-to-fine depth is not implemented, and the predicted uncertainty is not calibrated against any external reference, so it orders pixels by confidence without carrying a probabilistic guarantee.

9.2. Required Experiments and Metrics

Table 7 specifies eight runs that would convert the present proxy evidence into a defensible benchmark. Seven need no new collection — only the existing sequence, the frozen checkpoints and compute. P1 is the highest-value item: hiding one camera and reconstructing it from the remaining six yields an exact photometric ground truth, directly

testing the capability this protocol cannot establish. P7, the synchronized double rig, is the only route to a true Task C benchmark and the only one that cannot be started today.

The reporting set should extend along four axes: geometric accuracy as depth MAE and RMSE against held-out LiDAR, split into near, mid and far; semantic preservation as feature cosine similarity or mIoU; temporal consistency as optical-flow warping error rather than coverage variance; and downstream retention on an unmodified perception stack, which most directly answers fitness for purpose. Generated regions should stay labelled even when a stronger model improves their appearance: the provenance channel matters most exactly when the eye cannot find the completion.

10. Conclusions

We specified a provenance contract and a no-ground-truth evaluation protocol for cross-rig camera-data reuse, with a geometry-first pipeline implementing them on a calibrated seven-camera RGB-LiDAR platform. The design makes observability explicit: LiDAR and calibrated source RGB establish metric evidence, a shared SE(3) rig transform defines the target geometry, cycle and visibility gates reject unsupported correspondences, and residual completion is isolated behind a mask labelled generated and excluded from every geometric export.

The results characterize that protocol rather than certify the pipeline. The geometry branch reconstructs all seven target cameras under two parameterized rigs at 94.07% and 96.66% mean coverage, passes the display-completeness audit with no near-black component above 3 pixels, and holds coverage to a 0.44-point standard deviation across 20 consecutive frames of one camera under one rig. The learned branch, on 21 same-name small-baseline cases that are Task B rather than cross-rig synthesis, attains 98.24% mean observed coverage, recovers 44.90% of holes through the gate as a per-camera mean, 37.77% pooled over all hole pixels, confines generated pixels to 1.10% of the image, and modifies zero locked pixels in every case.

The boundaries of that evidence are set out in Sections 1.3 and 9.1 and softened nowhere. The operational one bears repeating: because we report no held-out geometric error and no downstream retention, **this paper does not demonstrate that the reprojected corpus is fit for training or validating a perception stack.** It contributes an auditable route to that determination.

Two findings are worth testing beyond this system: degradation under rig displacement is strongly camera-dependent, so a seven-camera mean can conceal a fivefold concentration of vehicle-region risk; and an acceptance specification written in terms of evidence is checkable, whereas a hole-free requirement is satisfiable by hallucination. Until a measured double-rig collection exists, the appropriate posture is the one adopted here: reconstruct what was observed, recover only what can be demonstrated, generate the remainder under an explicit label, and never let the third category be mistaken for the first.

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Institutional Review Board Statement: Not applicable. The study did not involve human participants or animals. It reuses a pre-existing road-collection sequence recorded by the data provider in public traffic; no recruitment, intervention or interaction with any person took place, and the authors collected no data themselves.

Informed Consent Statement: Not applicable. No human participants were enrolled. Incidental road users appear in the source imagery as they would to any road observer; licence plates, faces and sensitive geographic information are anonymized to the provider's requirements before any release, and neither the raw imagery nor any identifiable derived data is redistributed by the authors, as stated in Section 8.2.

Data Availability Statement: The inference code, the derived evaluation results, the manuscript sources and the audit scripts that produce every number reported here are openly available at <https://github.com/JHou-TongJi/cross-rig-provenance-nvs>. That repository contains the evaluation reports, provenance masks, configuration files and lineage records. The underlying road-collection RGB imagery, LiDAR point clouds and manufacturer calibration files were supplied under a challenge data-use agreement and are not redistributed by the authors; access requests should be directed to the data provider. Model checkpoints are referenced through controlled paths and are available from the corresponding author on reasonable request, subject to the licence terms of the third-party components listed in Table 4. The consequence for reproducibility is stated in Section 8.1 and repeated here so the two cannot be read apart: the repository reproduces the geometry-branch results given the sequence and the calibration, because that branch carries no learned component, whereas the learned-branch results are a non-reproducible demonstration, since the training data, losses, optimizer settings and checkpoint-selection splits behind the frozen weights cannot be released.

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